Networking: Concepts

INTRODUCTION

Initially computers were being used as ‘stand-alone’ systems to fulfill the organizational requirements. The concept of information sharing and services brought the computer networks into existence on which we are vastly dependent.

ARPANET (Advanced Research Project Agency Network) was the first network that used technique of packet switching in 1970

Communication Networks

The communications networks can be grouped into the following three categories based on the technology and communication media used by them:

i. **Public Switched Telephone Network:**

PSTN are managed by common carriers usually telephone companies / departments the world over.

The normal dial-up connections to the subscribers and permanent leased connections between two subscribers' points.

ii. **Public Switched Data Network**: Analogous to the public telephone network, many domestic common carriers provide data communications services via a specialized network called a Public Switched Data Network (PSDN). Most Internet Service Providers (ISPs) use combination of PDN and PSTN for providing Internet connectivity.

The permanent leased connections provide single traceable line between two subscriber points. The leased lines are generally less prone to noise than dial-up lines.

iii. **Integrated Services Digital Networks (ISDN)**: An ISDN is a network that provides end-to-end digital connectivity to support a wide range of services, including voice and non-voice services.

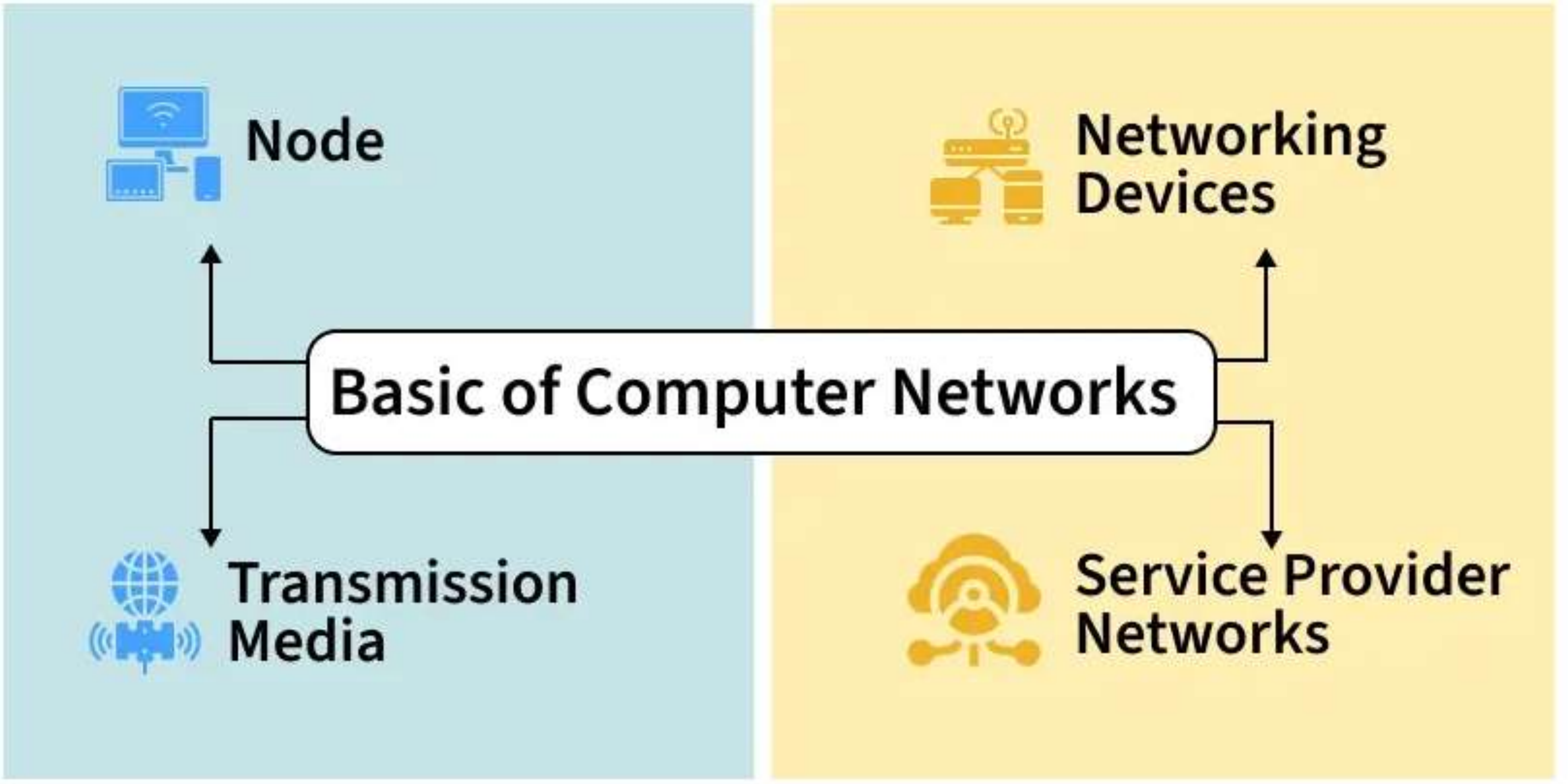
LAN-WAN-LAN- Intranetwork or Internet

Computer Networks

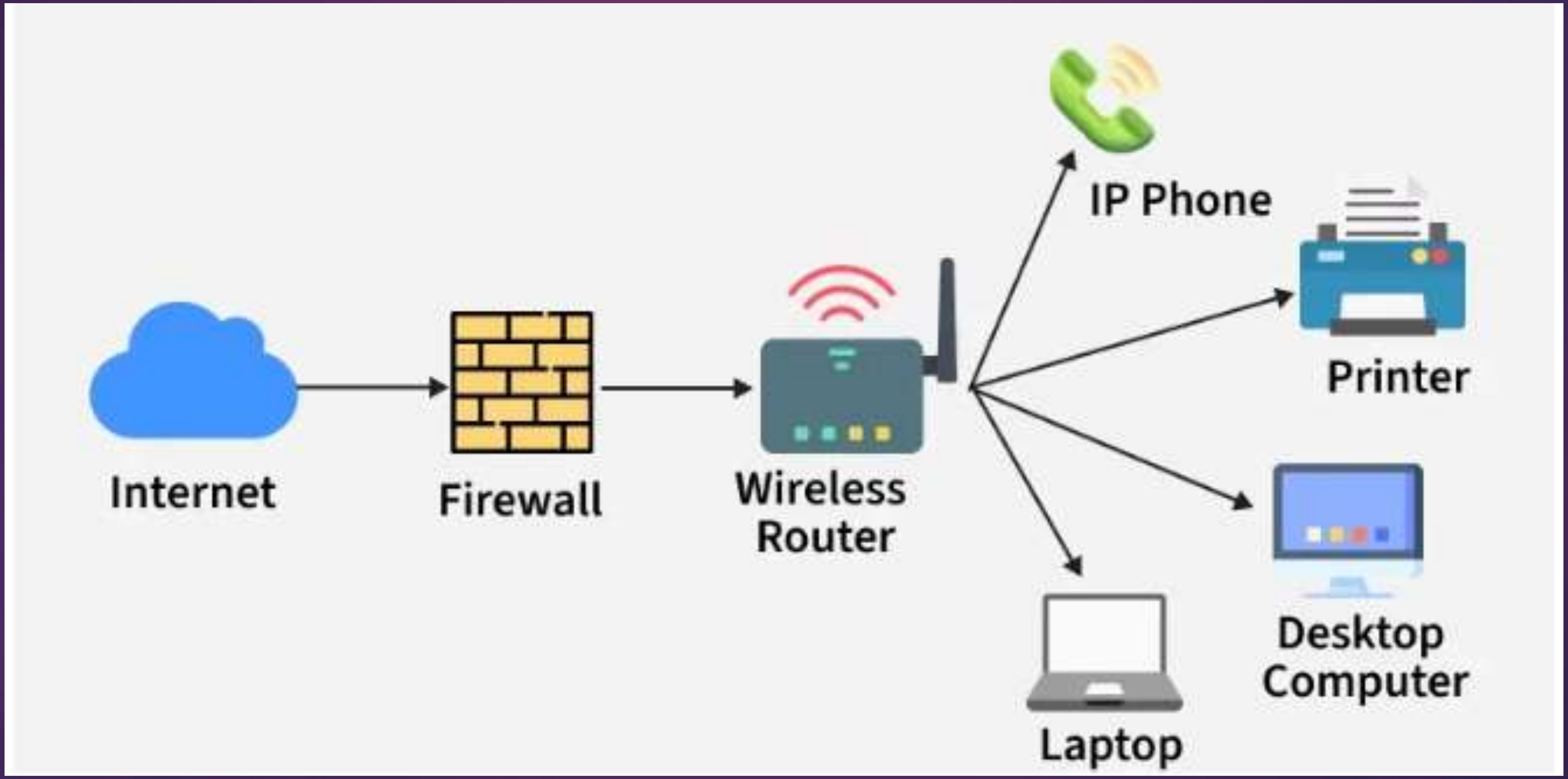
A computer communication network is an interconnection of a collection of several computers from which the user can select the service required and communicate with any computer as a local user.

Developments in communication technology have made it possible to interconnect geographically dispersed computing resources of different kinds and makes.

- Nodes are physical devices such as computers, mobiles, or printers.
- Routers and switches control the flow of information.
- Transmission media carry data from one device to another.
- Wired media include Ethernet and optical fiber cables.



- Computer network operates by enabling devices to communicate and exchange data using a shared communication system. Each device in the network follows predefined rules to ensure that data is transmitted accurately, efficiently, and securely.



- A network consists of nodes such as computers, servers, routers, and switches that send or receive data.
- These nodes are connected through links, which may be wired (cables, optical fiber) or wireless (Wi-Fi, radio signals).
- When data is sent, it is broken into small packets and transmitted across the network.
- Protocols define how data packets are formatted, transmitted, received, and acknowledged.
- Each device is identified by a unique IP address, which ensures data reaches the correct destination.
- Network devices like switches and routers forward data packets along the best available path.
- Security mechanisms, such as firewalls, monitor traffic and allow or block data based on security rules.

Goals and Uses

- **Resource Sharing:** Allow multiple users to share hardware, software, and data resources efficiently.
- **Internet and Cloud Access:** Enable access to the Internet, online services, and cloud-based applications.
- **Cost Efficiency:** Reduce operational and infrastructure costs through shared resources and centralized systems.
- **Reliability and Availability:** Improve system reliability using backup paths and fault-tolerant mechanisms.
- **Scalability and Growth:** Support easy expansion by adding new devices and services as demand increases.
- **Security and Control:** Protect data and network resources using authentication, access control, and monitoring.

Types of Computer Network Architecture

- **Client-Server Architecture:** Represents a type of computer network architecture in which nodes function as servers or clients, where the server manages client behaviour, known as Client-Server Architecture.

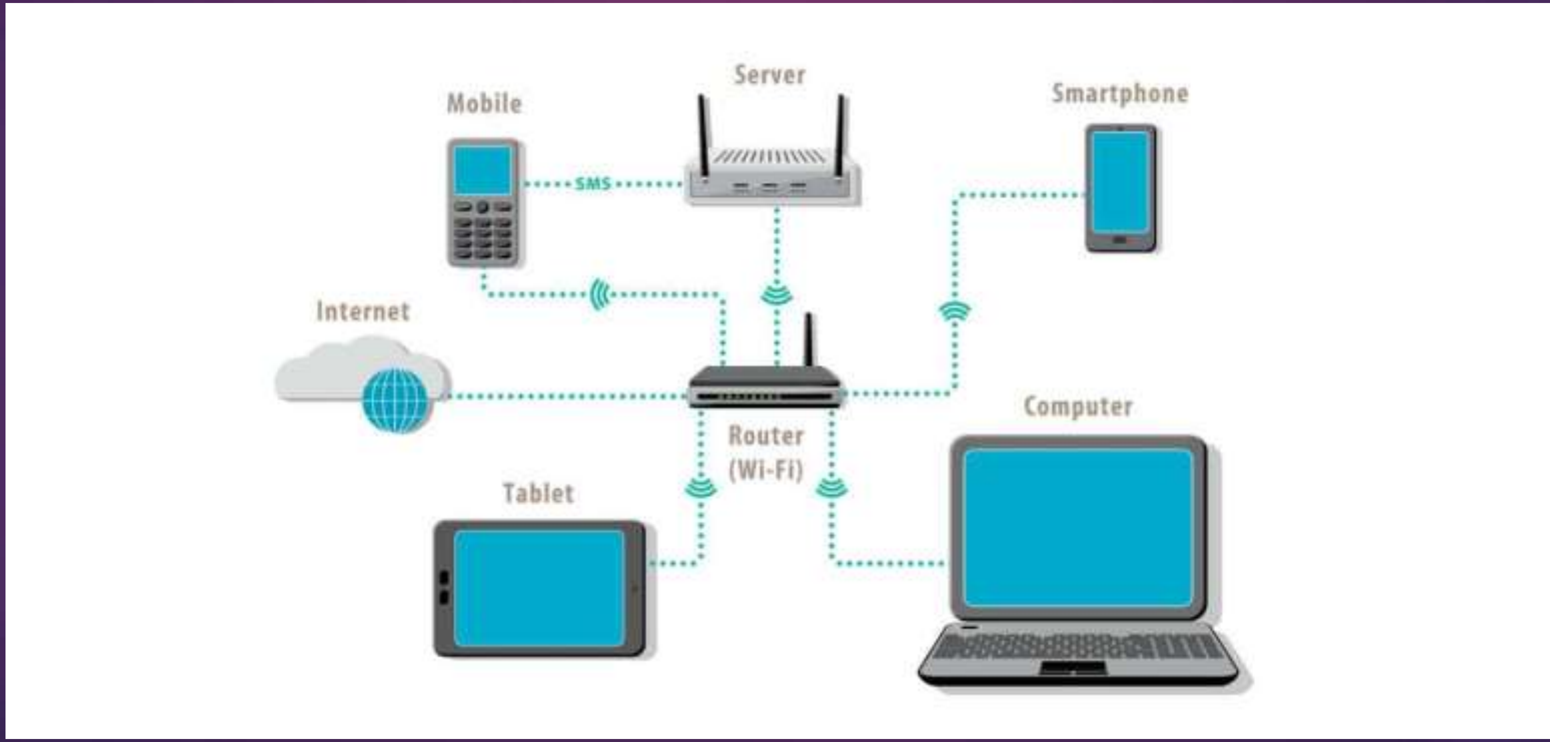
- **Peer-to-Peer Architecture:** Operates without any central server, allowing each device to act as either a client or a server, known as P2P (Peer-to-Peer) Architecture.

Types of Network

- PAN
- LAN
- MAN
- WAN

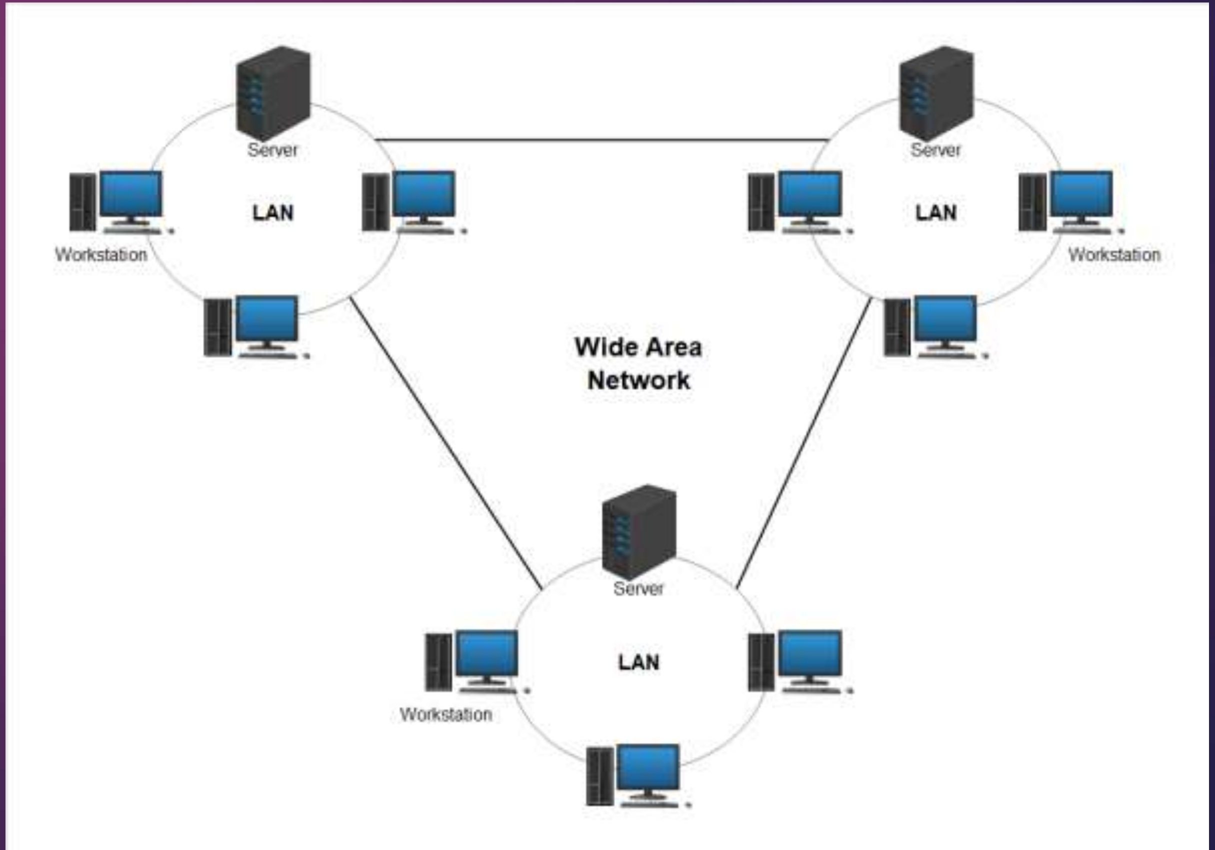
Local Area Network (LAN):

Interconnection of many computers within a given local area, more often premises of a single organization building. A very high speed of data transmission can be attained within a limited geographic area. LANs are typically configured in a star, bus or ring. Low speed LANs use telephone wires or copper cables, optical fiber cables are used to achieve high- speed transmission of data.



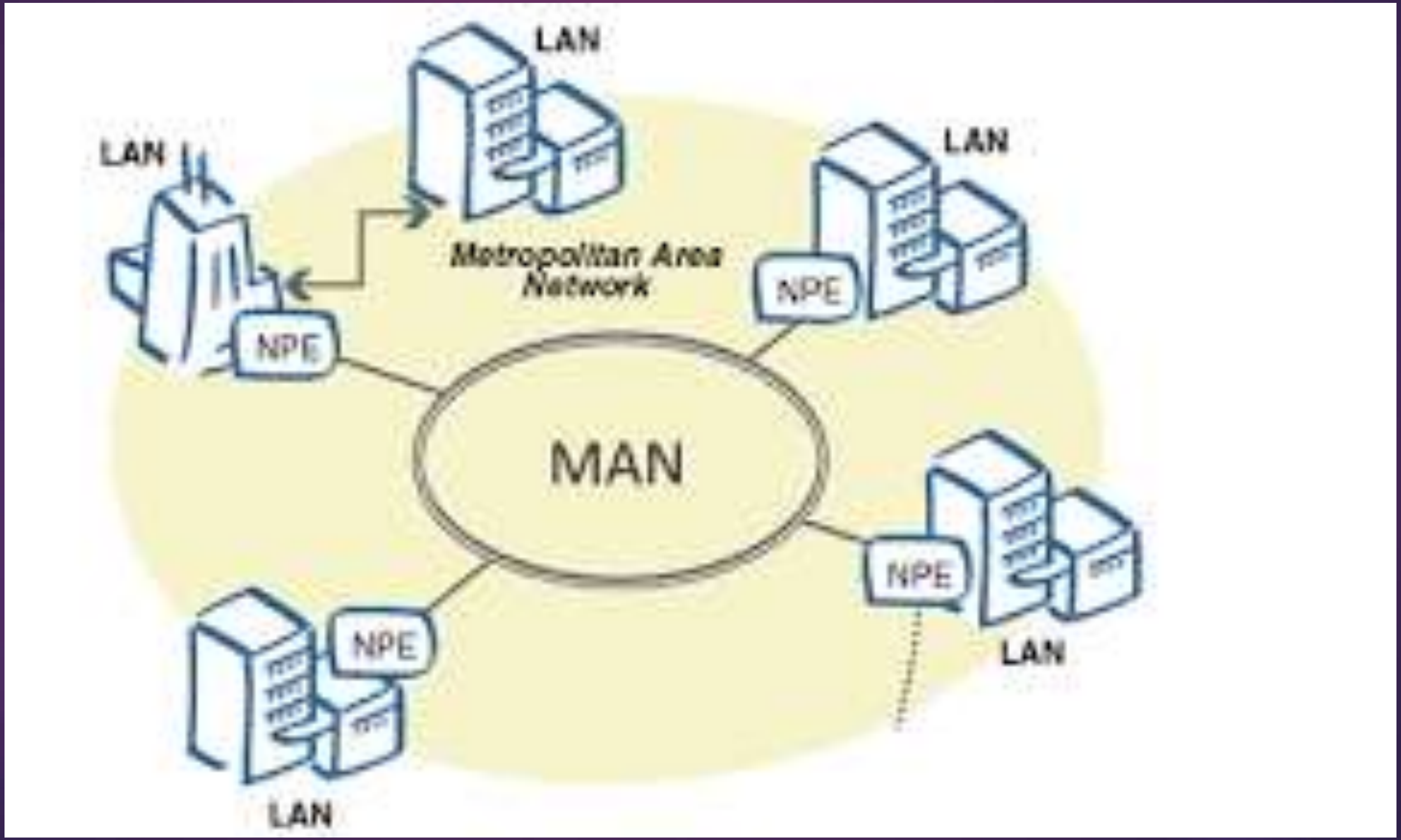
Wide Area Network (WAN)

WAN is used to interconnect a number of widely dispersed computers in various cities of a country or different countries. WANs use communication media maintained by telegraph or telephone companies. These networks usually have land telephone lines, underground coaxial cables, microwave communication and satellite communications.

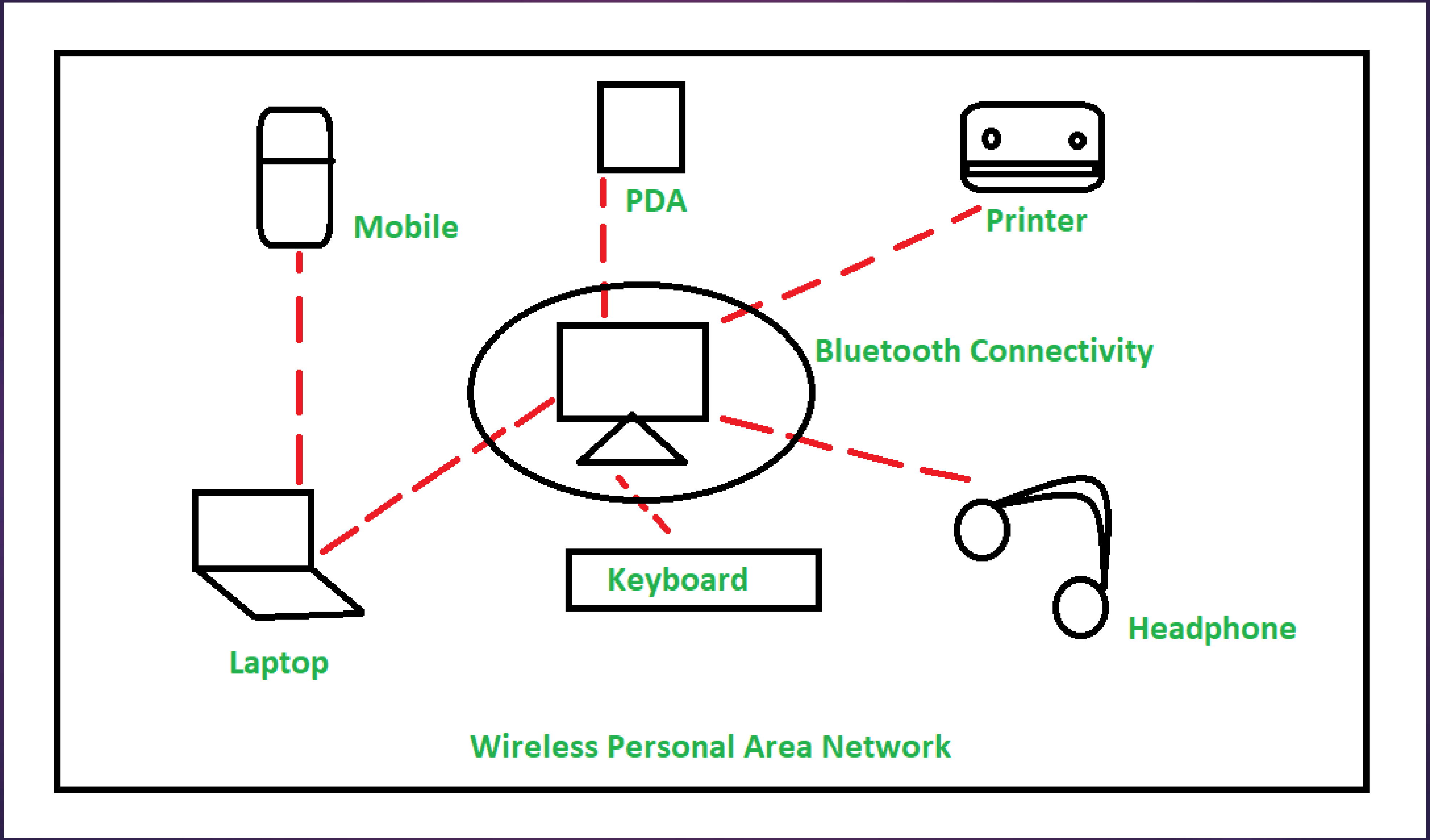


Metropolitan Area Network (MAN)

MAN refers to inter- connection within geographical limits of a city or town.



PAN


Mobile
PDA
Printer
Bluetooth Connectivity
Laptop
Keyboard
Headphone
Wireless Personal Area Network

Classification Based on Transmission Medium

1. **Broadcast Network**

This uses a shared communication medium among all connected devices.

- Data transmitted by one node is received by all other nodes in the network.
- Each data packet contains a destination address, and only the intended device processes it.
- Communication follows a one-to-many model.
- No dedicated physical path exists between sender and receiver.
- **Examples:**
- Ethernet (bus-based)
- Wireless LAN (Wi-Fi)

2. Point-to-Point Network

This uses dedicated links between communicating devices.

- Data is transmitted directly from one node to another.
- Communication follows a one-to-one model.
- Data may pass through intermediate devices (routers or switches).
- Requires routing algorithms to determine the best path.
- **Examples:**
- Internet
- Leased line networks
- Telephone networks

Classification Based on Ownership and Access Control

1. Private Network

This type of network is owned and managed by an individual, organization, or institution

- Access is restricted to authorized users only
- Requires authentication such as usernames, passwords, VPNs, or access tokens
- Used mainly for internal communication
- Offers higher security, privacy, and control
- **Examples:**
- Corporate intranet
- Office LAN
- Private cloud network

2. Public Network

This type of network is owned by service providers or government authorities

- Accessible to the general public
- Requires little or no authorization to connect
- Offers wide accessibility but lower security
- Usually shared by a large number of users
- Can operate over both wired and wireless media
- **Examples:**
- Internet
- Public Wi-Fi networks
- Mobile cellular networks

3. Hybrid Network

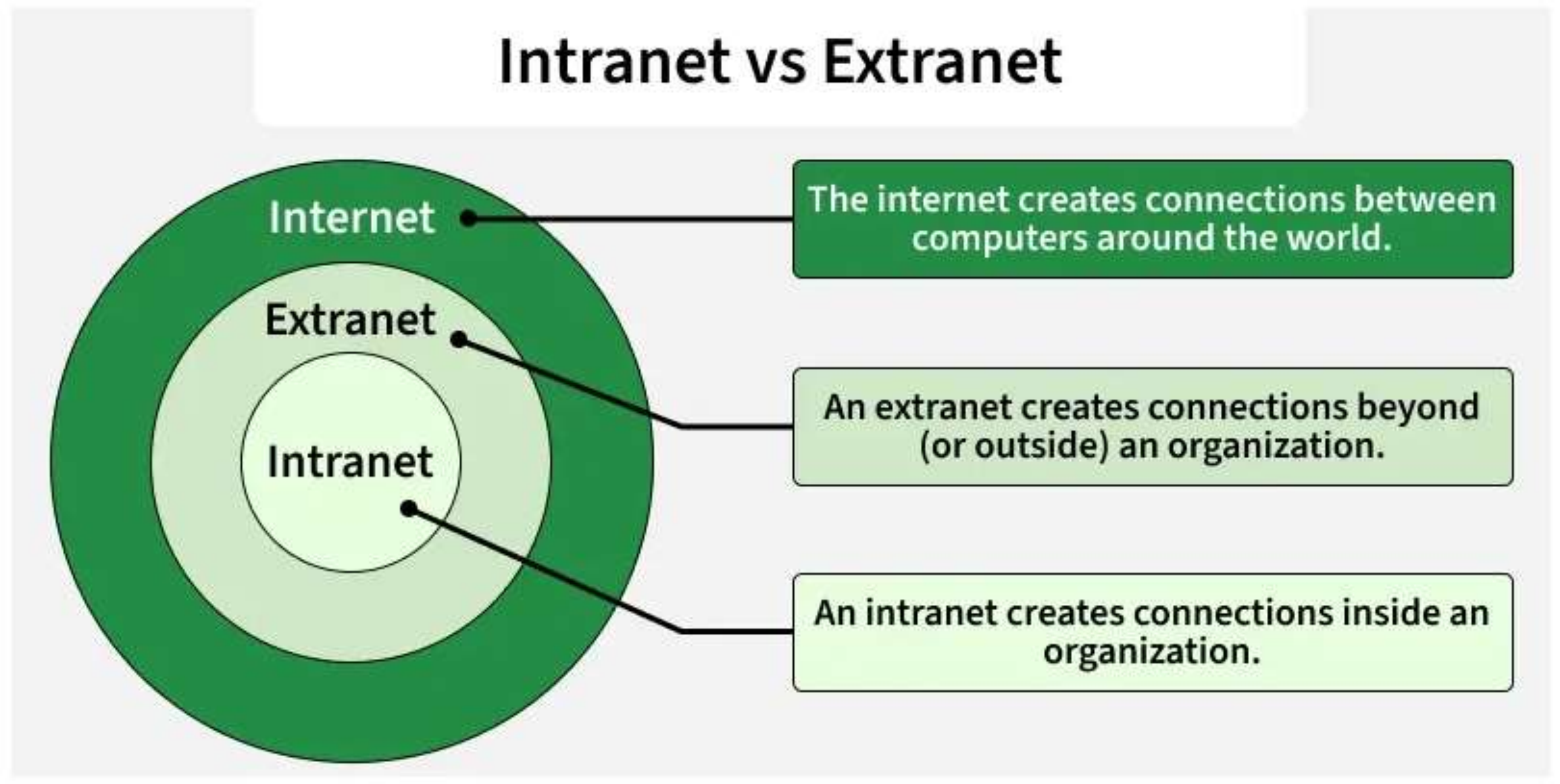
This type of network is combination of private and public networks

- Some parts of the network are restricted, while others are publicly accessible
- Allows organizations to maintain internal security while providing external connectivity
- Common in modern enterprise and cloud-based systems
- **Examples:**
- Corporate network connected to the Internet
- Enterprise cloud networks
- Online banking systems

Internetwork

An internetwork is a collection of two or more independent networks connected together to function as a single logical network.

- It enables communication between devices across different networks.
- The Internet itself is the largest example of an internetwork.
- Consists of multiple networks, not just individual computers.
- Uses networking devices to interconnect networks


Intranet vs Extranet
Internet
Extranet
Intranet
The internet creates connections between computers around the world.
An extranet creates connections beyond (or outside) an organization.
An intranet creates connections inside an organization.

1. Intranet

An intranet is a private network used within an organization. It is accessible only to authorized users such as employees.

- Owned and controlled by a single organization
- Access restricted using login credentials
- Used for internal communication and collaboration
- Improves information sharing within the organization

Uses

- Employee portals
- Internal emails and messaging
- Sharing company policies and documents
- Internal applications (HR, payroll, project management)

Examples

- Corporate internal website
- University campus portal
- Government department internal systems

2. Extranet

An extranet is an extension of an intranet that allows limited access to external users. it connects the organization's internal network with trusted outsiders.

- Partially private network
- Access provided to partners, vendors, suppliers, or customers
- Requires authentication and authorization
- Often implemented using VPN or secure web access

Uses

- Business-to-business (B2B) communication
- Supply chain management
- Customer portals
- Partner collaboration platforms

Examples

- Supplier management system
- Online banking portals
- Vendor access portals

Internet vs. World Wide Web

Many people use the Internet and World Wide Web (WWW) interchangeably, but they are not the same.

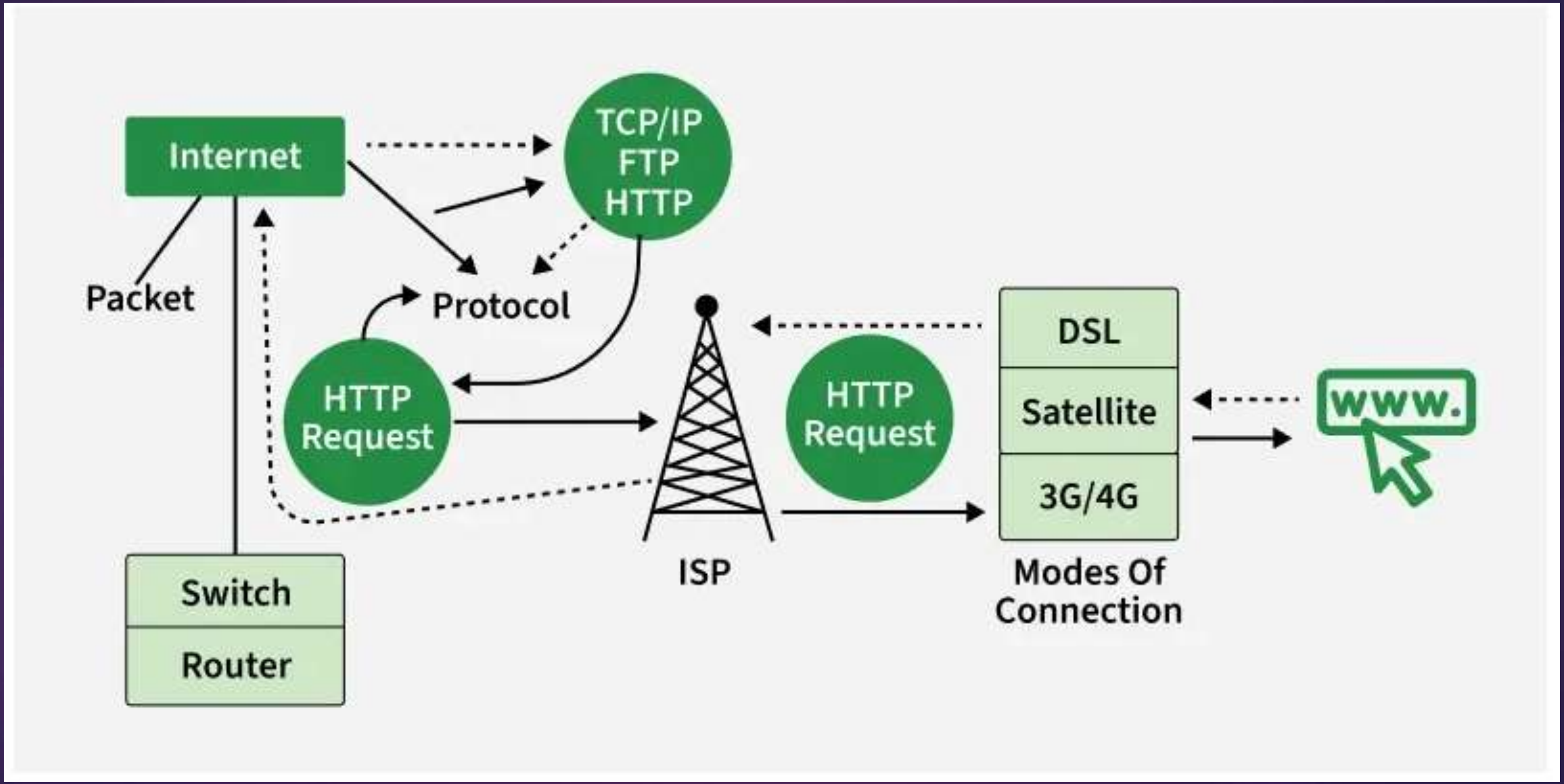
- The Internet is the infrastructure — the physical network of cables, satellites, and routers.
- The World Wide Web is one of the services that runs on the Internet — it's where websites and web pages exist, accessible through browsers like Chrome or Safari.

WWW: 3 key technologies

In 1989–1991, Sir Tim Berners-Lee, a British scientist, invented the World Wide Web

- **HTML (HyperText Markup Language)** – for creating web pages
- **HTTP (HyperText Transfer Protocol)** – for communication between browsers and servers
- **URLs (Uniform Resource Locators)** – for addressing pages on the Web

Working of Internet



Protocols – The Rules of the Road

All devices on the Internet follow a set of communication rules called protocols. The most important ones are:

- **TCP (Transmission Control Protocol)** – Breaks data into smaller packets and ensures they arrive safely.
- **IP (Internet Protocol)** – Assigns addresses and delivers packets to the correct destination.
- **HTTP/HTTPS** – Used for websites.
- **SMTP** – Used for sending emails.
- These protocols ensure smooth, reliable, and secure communication worldwide.

IP Addresses – The Street Address

Every device connected to the Internet has a unique IP Address (e.g., 192.168.1.1).

It's like a home address that tells data packets where to go and where they came from. Modern Internet systems are transitioning from IPv4 to IPv6 to handle the massive number of connected devices.

Data Packets – The Small Boxes

- When you send information — like a photo or an email — it's divided into small chunks called packets. Each packet travels through the Internet independently, sometimes by different routes, and they are reassembled at the destination into the original file.

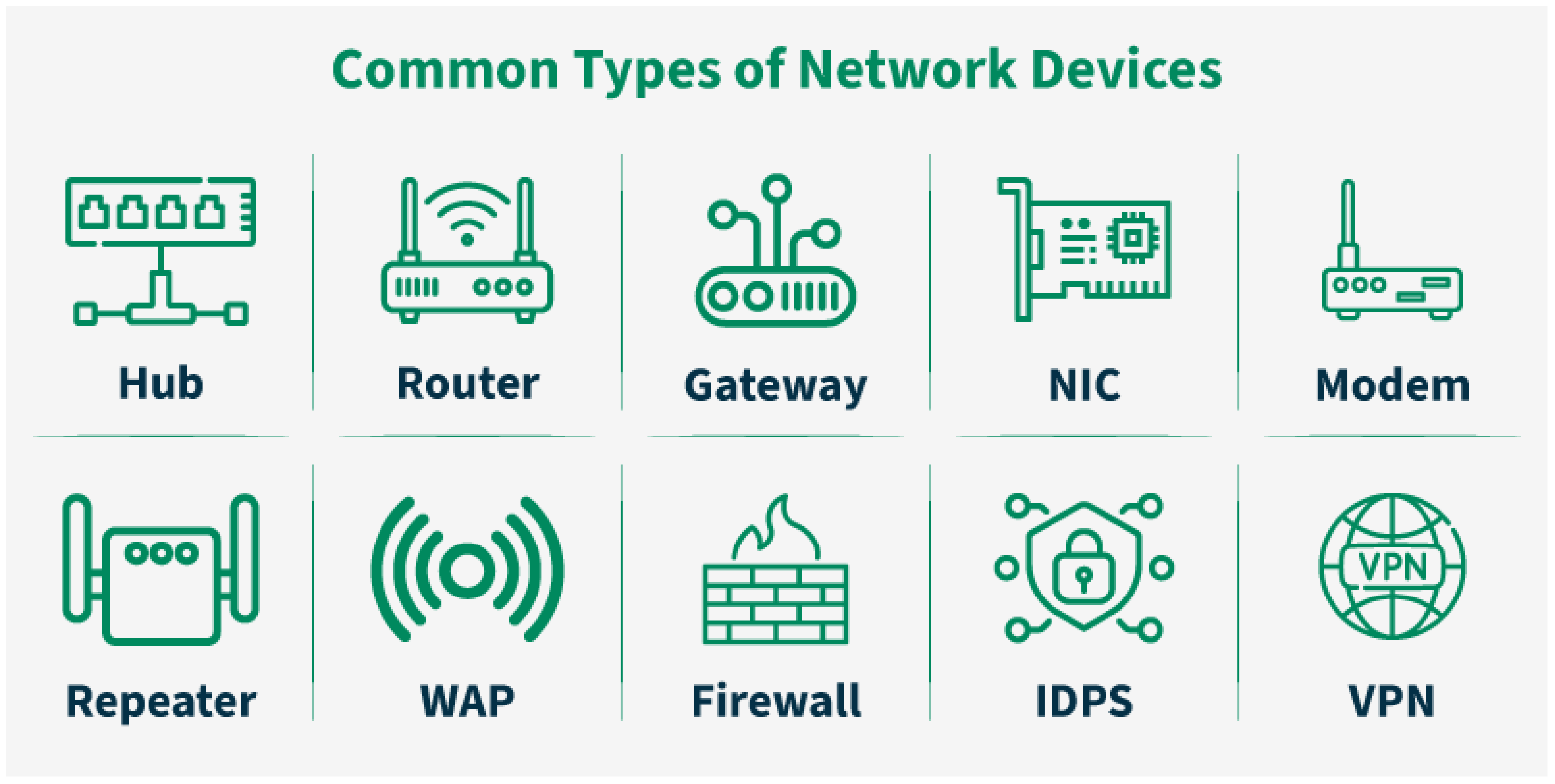
Internet Components

- **Routers:** Direct packets along the most efficient path.
- **Servers:** Store and deliver data like websites, emails, and videos.
- **ISPs (Internet Service Providers):** Companies like Airtel, Jio, AT&T, or Verizon that connect users to the Internet.
- **DNS (Domain Name System):** Translates website names (like google.com) into IP addresses.

Basic Components of Computer Networks

a. Network hardware

The basic component of computer network hardware is a computer. Computers on a network can be divided into two categories, server and clients or nodes. Server is the computer of higher power, and speed. It costs more. To this computer resources are attached. And the clients, also known as nodes access, are the r esources which are attached to server. In peer to peer computer networks there are no servers.


Common Types of Network Devices
Hub
Router
Gateway
NIC
Modem
Repeater
WAP
Firewall
IDPS
VPN

- Network Devices are the physical appliances required for communication and interaction between computers on a computer network.
- Enable communication by transmitting and receiving data between devices.
- Allow devices to connect to networks efficiently and securely.
- Improve network performance by reducing congestion and managing traffic.
- Extend network coverage and solve signal loss or attenuation problems.

Layer 1 Devices: Physical Layer

1. Hub

- It is a central connection point for devices in a LAN.
- It takes an incoming signal on one port and blindly broadcasts it to all other ports without knowing the recipient.
- High network traffic collisions and security risks (everyone sees everyone's data).
- It is a obsolete device, replaced by Switches.

2. Repeater

- It regenerates signals to extend the range of a network.
- It receives a weak signal (due to attenuation over long cables), amplifies it, and retransmits it.
- Extending WiFi range or Ethernet cables beyond 100 meters.

- **3. Modem (Modulator-Demodulator)**
- This bridges the gap between digital computer data and analog signals, allowing devices to connect to the internet via telephone, cable, or fiber lines.
- Connects your home network to the ISP (Internet Service Provider).
- Converts digital signals from your computer into analog signals for telephone/cable lines (Modulation) and vice-versa (Demodulation).

Layer 2 Devices: Data Link Layer

These devices work with MAC Addresses (Physical Addresses). They are smarter than hubs.

4. Switch

- A switch is a high-speed networking device that connects devices (computers, printers, servers) within a Local Area Network (LAN),
- Unlike hub, switch learns the MAC address of every connected device.
- It sends data only to the specific port where the destination device is connected.
- Reduces collisions and improves security and speed.

5. Bridge

- This connects two separate LAN segments to make them appear as one.
- Filters traffic based on MAC addresses to keep local traffic local, reducing congestion.
- Largely replaced by Switches (which are essentially multi-port bridges).

6. Access Point (AP)

- It is a networking hardware device that allows Wi-Fi-enabled devices, such as laptops and smartphones, to connect to a wired network.
- Acts as a bridge between wired Ethernet and wireless Wi-Fi devices.
- An AP only provides Wi-Fi signal and it does not route traffic or assign IP addresses.

Layer 3 Devices: Network Layer

These devices work with IP Addresses (Logical Addresses) and connect different networks together.

7. Router

- It is a networking device that connects multiple networks and directs data packets between them using IP addresses.
- It uses IP Addresses to determine the best path for data packets to travel.
- It maintains a Routing Table to make these decisions.
- Routers stop broadcast traffic, isolating networks from each other.

8. Brouter (Bridging Router)

- It is a networking device combining bridge and router functions, operating at both the Data Link and Network layers of the OSI model.
- It can route known protocols (like TCP/IP).
- It bridge unknown protocols (like non-routable legacy traffic) at the same time and rarely used today.

Layer 4-7 Devices: Advanced Processing

9. Gateway

- It is a hardware device or software node that acts as an entry/exit point between two distinct networks using different protocols, translating data to ensure compatibility.
- It can convert protocols, data formats, or architectures (e.g., connecting a TCP/IP network to a legacy Mainframe network).
- An Internet Gateway translates your private LAN requests into public internet requests.

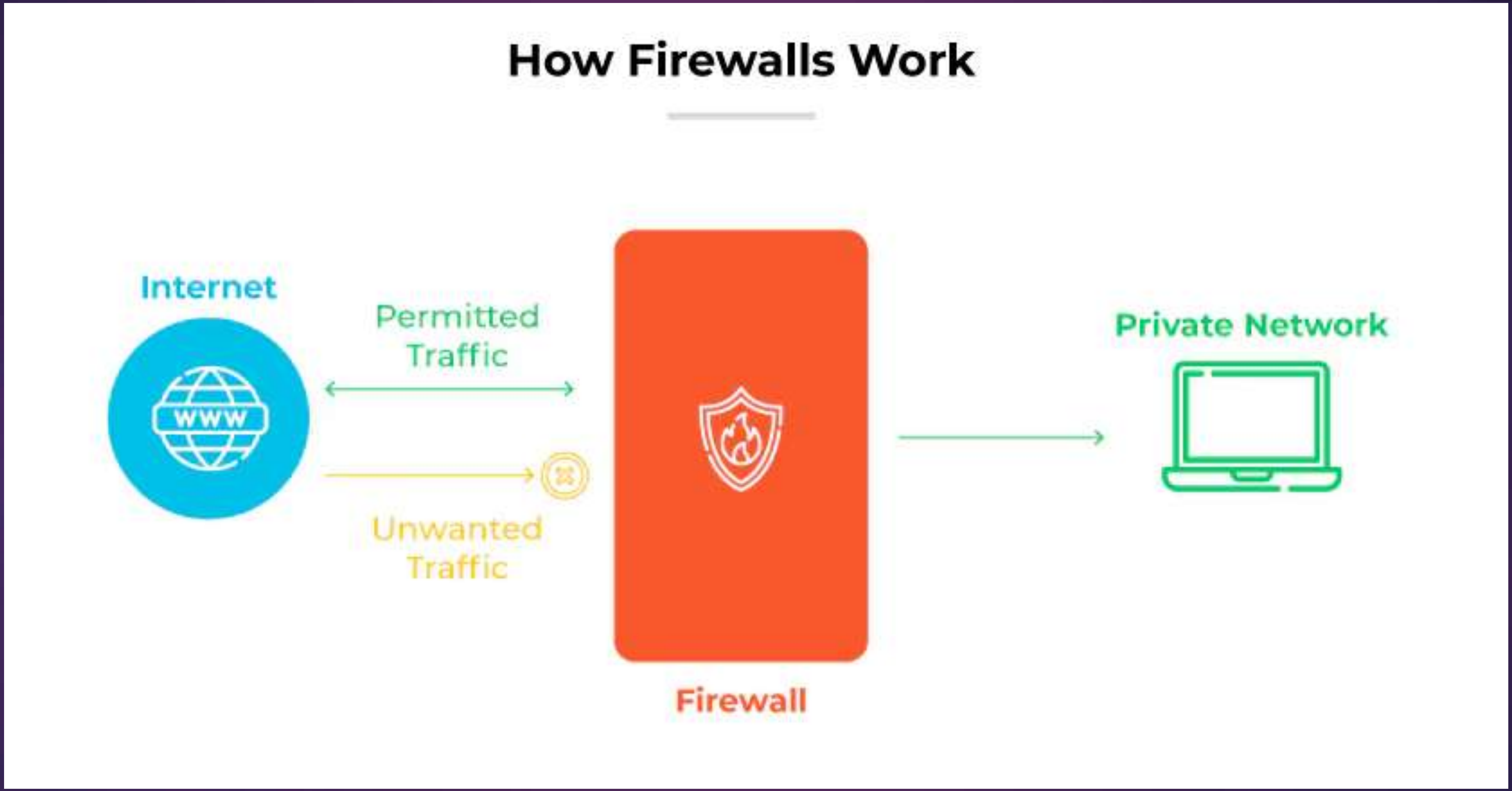
10. Firewall IDS IPS

Firewall

- A firewall is a network security solution that inspects and regulates traffic based on predetermined security rules, allowing, denying, or rejecting the traffic accordingly.
- **checkpoint** between internal networks and potential external threats
- analyze data packets against defined **security protocols**

- Data packets, which are structured for internet transit, carry essential information

  including their source, destination, and other crucial data that defines their journey across the network


How Firewalls Work
Internet
www
Permitted
Traffic
Unwanted
Traffic
Firewall
Private Network

Intrusion Detection System

- identifies potential threats and weaknesses in networked systems.
- An IDS examines network traffic, alerting administrators to suspicious activities without intervening in data transmission.

- IDSes are positioned out of the main traffic flow.- Mirroring

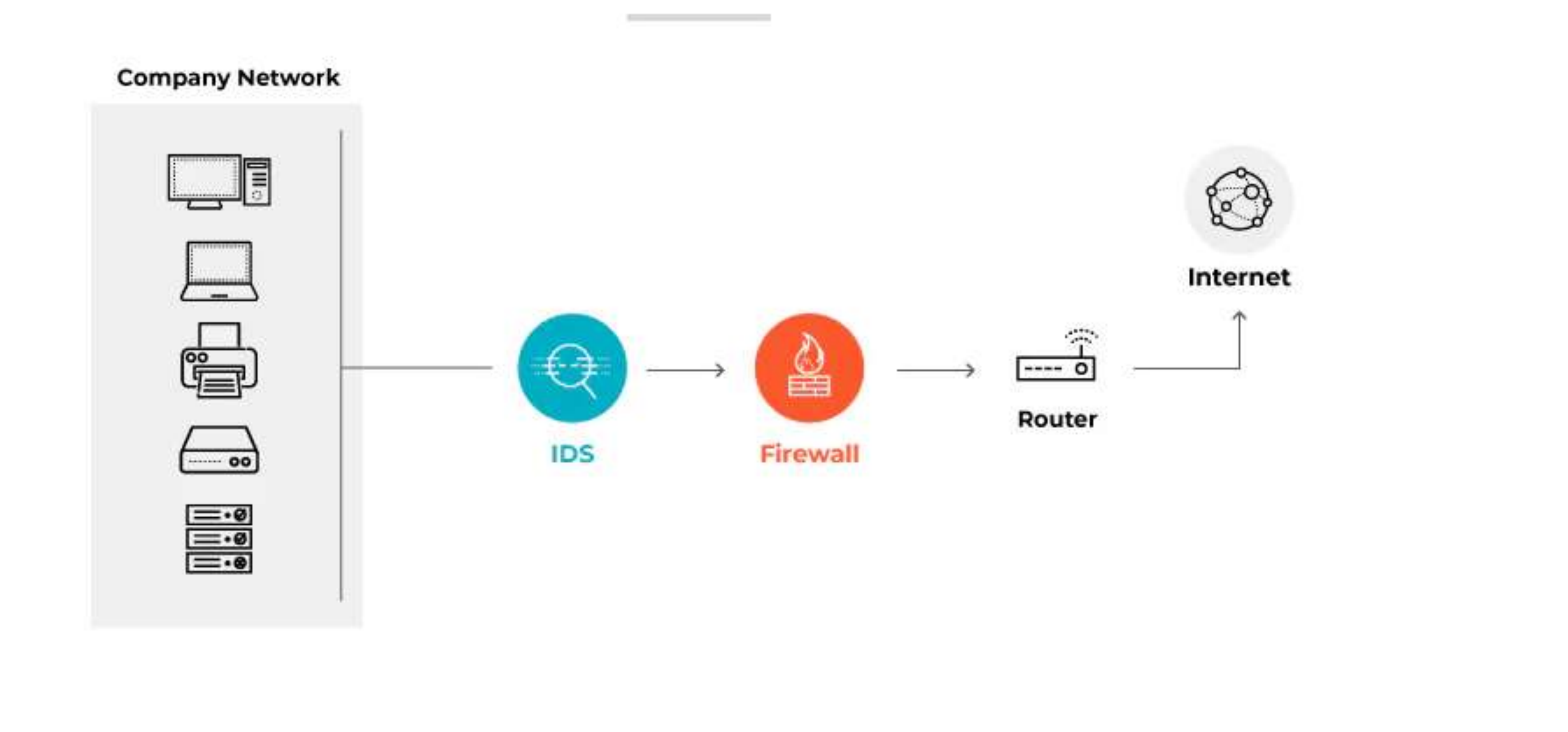
- IDS systems come in various forms

network intrusion detection system (NIDS), host based intrusion detection system (HIDS), protocol based (PIDS), application protocol based (APIDS), and hybrid

The two most common variations are signature based IDS and anomaly based IDS

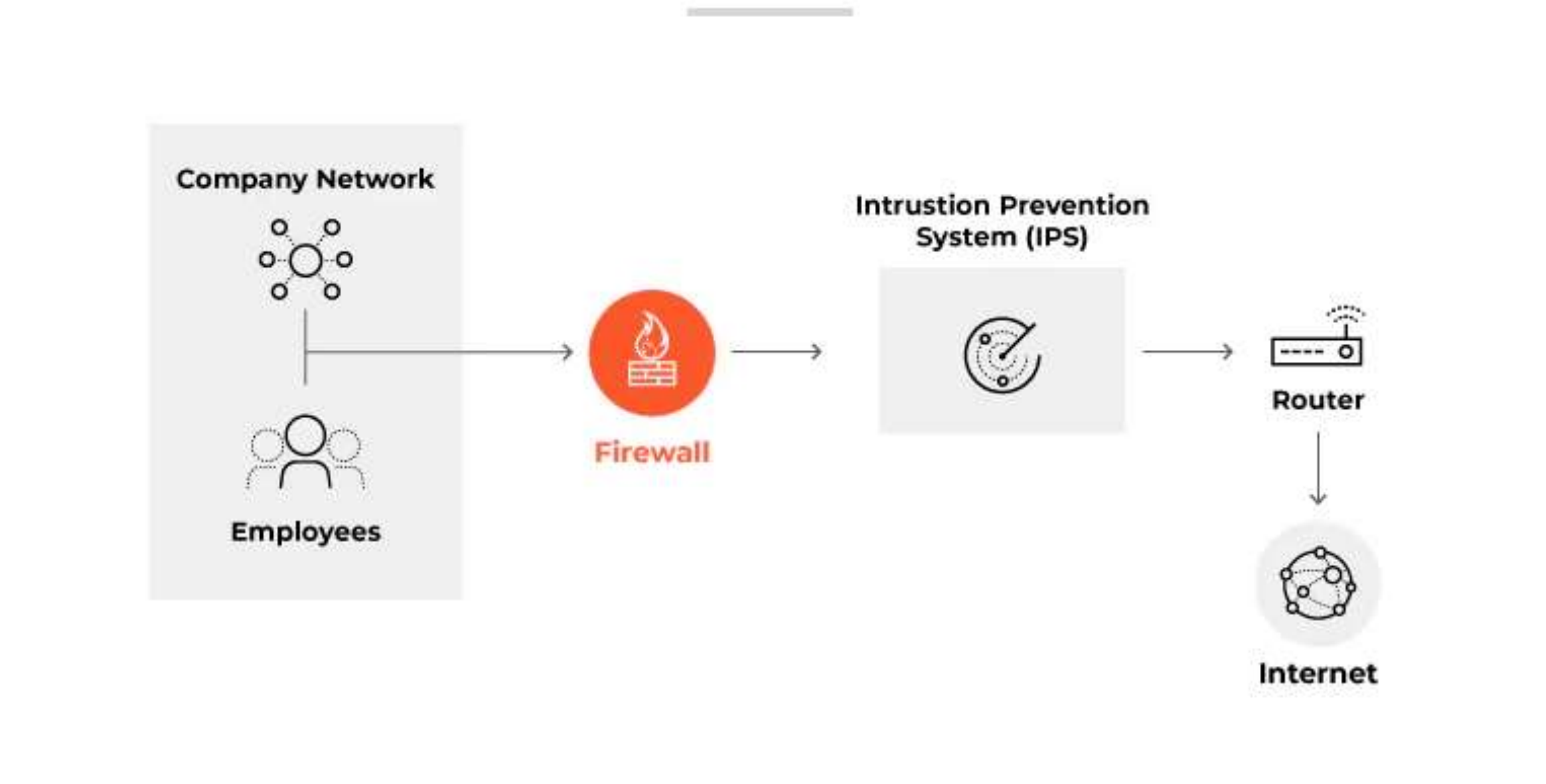
Hub vs. Switch vs. Router

Feature	Hub	Switch	Router
Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	None (Bits)	MAC Address	IP Address
Data Flow	Broadcast (to all)	Unicast (to destination)	Route (best path)
Intelligence	Dumb	Smart	Smarter
Used For	Obsolete	Connecting Devices (LAN)	Connecting Networks (WAN)


Company Network
IDS
Firewall
Router
Internet

Intrusion Prevention System

- dynamic security solutions that intercept and analyze malicious traffic.
- They operate preemptively to mitigate threats before they can infiltrate network defenses. This reduces the workload of security teams.
- IPS placement is in the direct path of network traffic
- IPS systems can signal alerts, discard harmful data, block source addresses, and reset connections to prevent further attacks.


Company Network
Employees
Firewall
Intrustion Prevention
System (IPS)
Router
Internet

Basic Components of Computer Networks

b. Transmission media

Communication of data propagation and processing of signals is called transmission. Signals travel from transmitter to receiver via a path. This path is called medium. Medium can be guided or unguided.

Guided Media

Unguided Media

Guided Media

In guided media, data is sent along a physical path i.e. cables. There are several types of cables used in network. The type of cable chosen for a network is related to the network's topology, protocol and size. Different types of cables are: coaxial cables, twisted pair copper wire, and optical fiber cable.

- Coaxial cable looks like cable that brings the cable TV signal to television.
- Twisted pair copper wire cable looks like phone cable. Twisted pair cables come in two varieties, shielded and unshielded.
- Optical fiber cable

Unguided Media

Here no wire is installed. The data communication is predominantly sent by radio waves and microwaves.

Characteristics of Transmission Media

Characteristics of Transmission Media

When selecting a communication channel, its effectiveness is measured by five critical performance metrics that dictate the efficiency and reliability of data flow.

1. Bandwidth(Throughput Capacity)

- Bandwidth represents the maximum data transfer rate a medium can support, typically measured in *bits per second (bps)*. In high-performance networking, media like fiber optics offer significantly higher bandwidth than copper, enabling the concurrent transmission of high-definition video and massive datasets without congestion.

2. Delay and Latency

Delays in a computer network refer to the time it takes for a data packet to travel from the sender to the receiver. This delay is caused by several factors at different stages of data transmission and is commonly called network latency

Main Types of Delays in Computer Networks:

1. Transmission Delay

It is the amount of time needed to send all the bits of a data packet from the sender onto the communication medium. It starts when the first bit is sent and ends when the last bit is transmitted onto the link. This delay depends on how fast the link can transmit data, not on how far the data travels.

Example: If a link has a bandwidth of 1 bps, it can transmit 1 bit per second. To transmit a packet of 20 bits, it will take 20 seconds, because only one bit is sent each second.

- *Transmission Delay (Tt) = L / B*

This delay depends upon the following factors:

- If there are multiple active sessions, the delay will become significant.
- Increasing bandwidth decreases transmission delay.
- MAC protocol largely influences the delay if the link is shared among multiple devices.
- Sending and receiving a packet involves a context switch in the operating system, which takes a finite time

2. Propagation Delay

It is the time taken by a data signal to travel through the transmission medium from the sender to the receiver after it has been transmitted. It starts when the first bit enters the medium and ends when that bit reaches the destination. This delay depends on the distance between the devices and the speed of signal propagation in the medium, not on the packet size or bandwidth.

Example: If a signal travels at a speed of 2×10^8 m/s and the distance between sender and receiver is 200 km, then the propagation delay is 0.001 seconds (1 ms).

Propagation Delay (Tp) = D / S

- *D = distance between sender and receiver (in meters)*
- *S = propagation speed of the signal (in meters per second)*

This delay depends upon the following factors:

- Distance: Greater distance increases propagation delay.
- Transmission Medium: Different media (fiber, copper, wireless) have different propagation speeds.
- Signal Speed: Higher propagation speed reduces delay.
- Physical Path: Longer or indirect paths increase propagation delay.

3. Queueing Delay

Queueing delay is the time a packet spends waiting in a queue (buffer) before it can be processed or transmitted. When a packet reaches a router or destination, it is not processed immediately if other packets are already waiting. The waiting time in the buffer is called queueing delay.

In practical networks, queueing delay varies based on traffic conditions. However, it can be mathematically modeled using queueing theory,

This delay depends upon the following factors:

- Queue Size: Larger queues can cause longer waiting times, while an empty queue results in little or no delay.
- Arrival Rate of Packets: If many packets arrive within a short time, queueing delay increases.
- Number of Servers/Links: Fewer processing servers or output links lead to higher queueing delay.

4. Processing Delay

Processing delay is the time taken by a network device (such as a router or switch) to process a received packet. This includes examining the packet header, checking for errors, updating fields like TTL, and determining the next forwarding path.

There is no fixed formula for processing delay because it depends on the speed and efficiency of the processor, which varies across devices.

This delay depends upon the following factors:

- Processor Speed: Faster processors result in lower processing delay.
- Packet Processing Complexity: More complex processing increases delay.

Total Delay

Total delay is the overall time taken by a data packet to travel from the sender to the receiver in a computer network. It is the sum of all delays experienced by the packet at different stages during transmission.

$$T_{total} = Tt + Tp + Tq + Tpro$$

Where:

- *Tt = Transmission Delay*
- *Tp = Propagation Delay*
- *Tq = Queueing Delay*
- *Tpro = Processing Delay*

If queueing and processing delays are negligible, then: $T_{total} = Tt + Tp$

3. Total Cost of Ownership (TCO)

- The cost factor encompasses not just the initial capital expenditure (CAPEX) for the physical media, but also the operational expenditure (OPEX). For example, while fiber-optic cable is more expensive to install than twisted-pair, its durability and low maintenance often result in a lower long-term TCO.

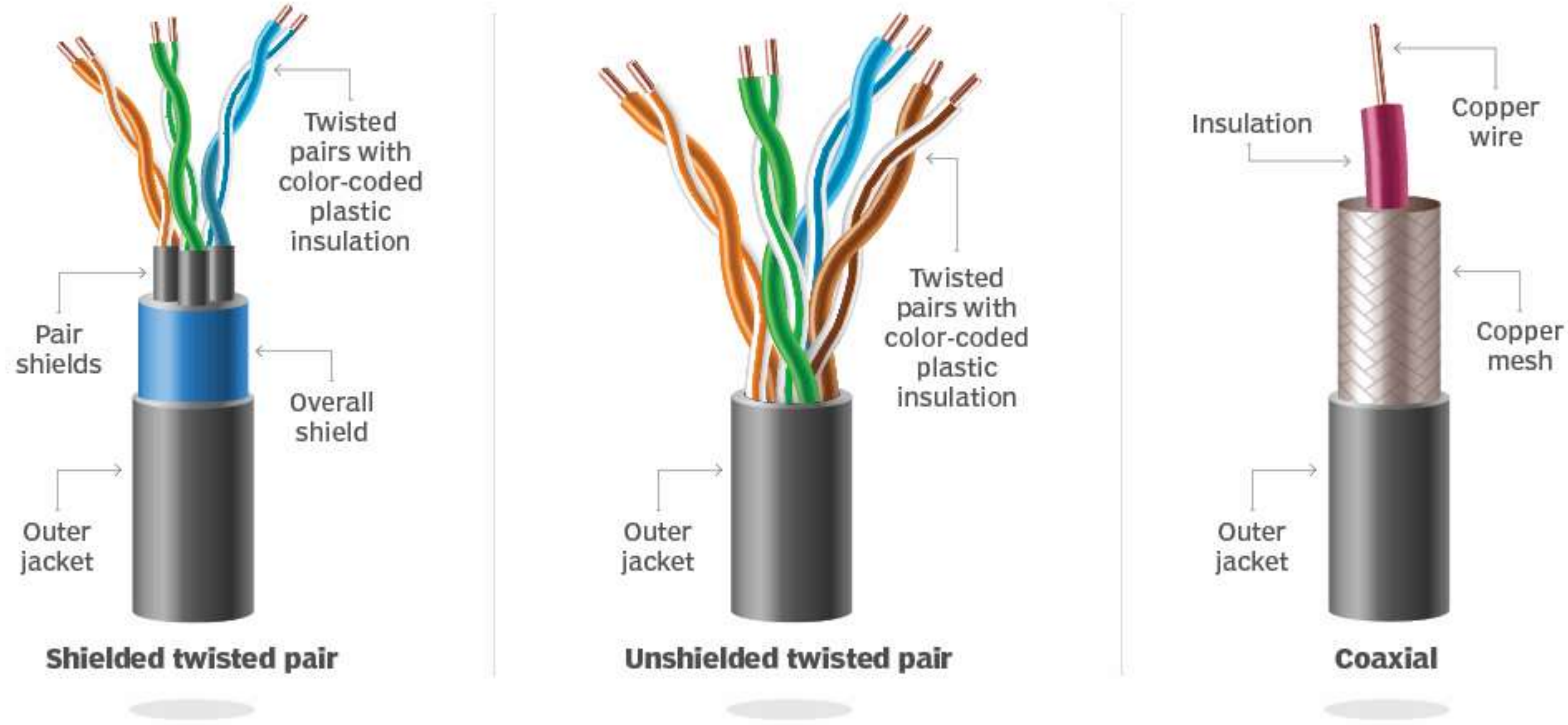
4. Attenuation and Transmission Distance

- Attenuation is the gradual loss of signal strength as it travels. Every medium has a specific maximum effective distance before requiring a repeater or amplifier. Fiber optics excels here, maintaining signal integrity over kilometers, whereas Ethernet cables (UTP) are generally limited to 100 meters.

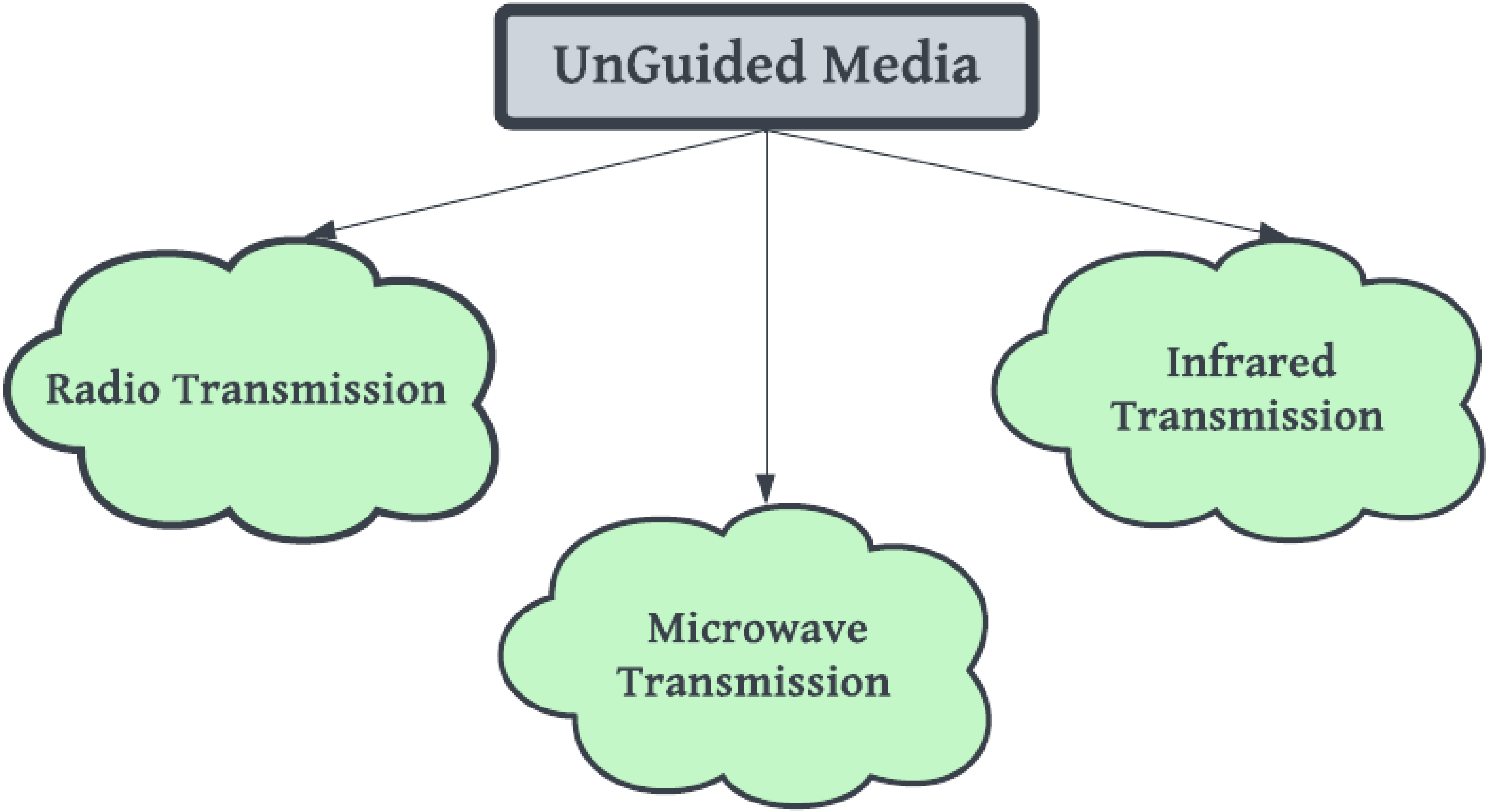
5. Signal Security and EMI Immunity

- Security measures how vulnerable a medium is to "eavesdropping" or Electromagnetic Interference (EMI). Wireless signals are susceptible to interception and noise, whereas fiber optics is virtually immune to EMI and extremely difficult to tap without detection, making it the most secure choice for sensitive backbones.

Enterprise cable options compared



ILLUSTRATIONS: ZUPERIA/GETTY IMAGES


UnGuided Media
Radio Transmission
Microwave Transmission
Infrared Transmission

Network Card

Most important part of connection is the network card. This is the middle part of connection. These cards of different bits. Each card has its own method of sending information (network protocol) through the cable. The most commonly used is **Ethernet Protocol**.

A network card is called Interface card, network adapter, a NIC etc. It is a circuit board or chip which allows the computer to communicate to other computers on a network.

Wired,Wireless

There are four basic types of modems for a PC: External, USB, Internal and Built-in.External and USB are set on to desk outside the PC, while as Internal and Built-in are inside the PC.

c. Network Operating Software

Network operating software (NOS) is a collection of software and associated protocols that allow a set of autonomous computers, which are interconnected by a computer network, to be used together in a convenient and cost effective manner. It is similar to any other operating system like windows, DOS, etc. except it operates over more than one computer.

It controls operation of the network system, including who uses it, when they can use it, what they have access to, and which network resources are available. At a basic level, the NOS allows network users to share files and peripherals such as disks and printers. They provide data integrity and security. The examples can be categories of NOS: The NetWare, LAN Manager, Solaris and Windows 2000 etc.

- **Peer to peer software Client Server Based (Two Tier)**

- **Peer to peer software**

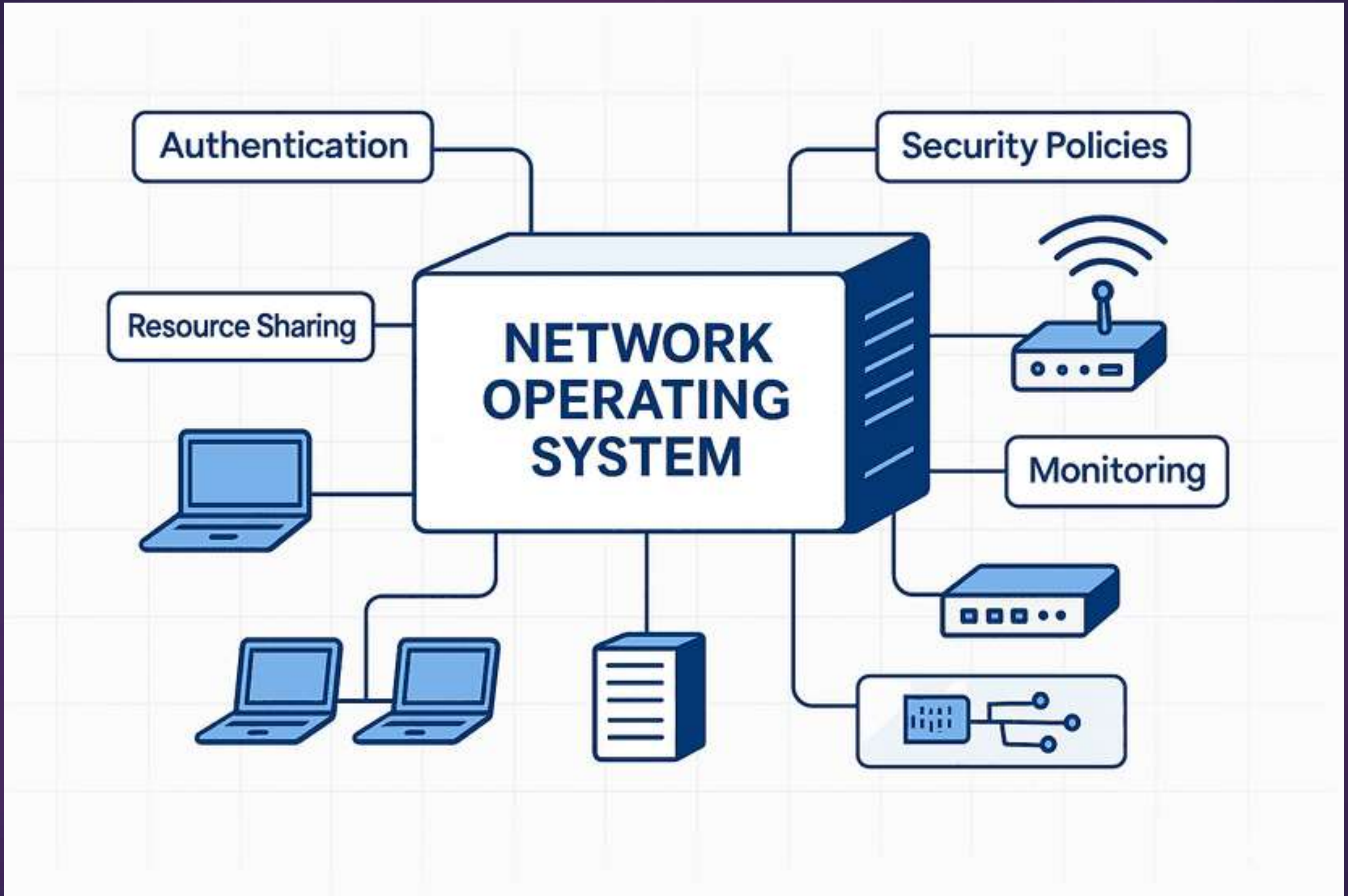
In peer to peer networking operating software users can share resources and files located on their computers and can access shared resources on other computers. There is no central server. All computers in the network are equal. They have similar capabilities and resources. Centralized control is impossible in this kind of architecture. The user of each machine is also the machine’s administrator.

- **Client Server Based (Two Tier)**

This software is in two parts. One part which includes functions and services resides in one or more exclusive (dedicated) computers. This part is called server. It provides security and access to resources. Another part called 'client' resides on other computers (nodes / client). They access resources on the server. The network operating system allows multiple users to simultaneously share the same resources irrespective of physical location.

Client Server Based (Three Tier) where a client-software is split into two parts. **Browser** – (user-interface), (thin client) and **Logic.** Thus two tier client-server becomes three tier architecture.

The logic which describes how to access and process data is moved to a new server. This new server is server for thin client. Nothing changes in the server side


Authentication
Security Policies
Resource Sharing
NETWORK OPERATING SYSTEM
Monitoring

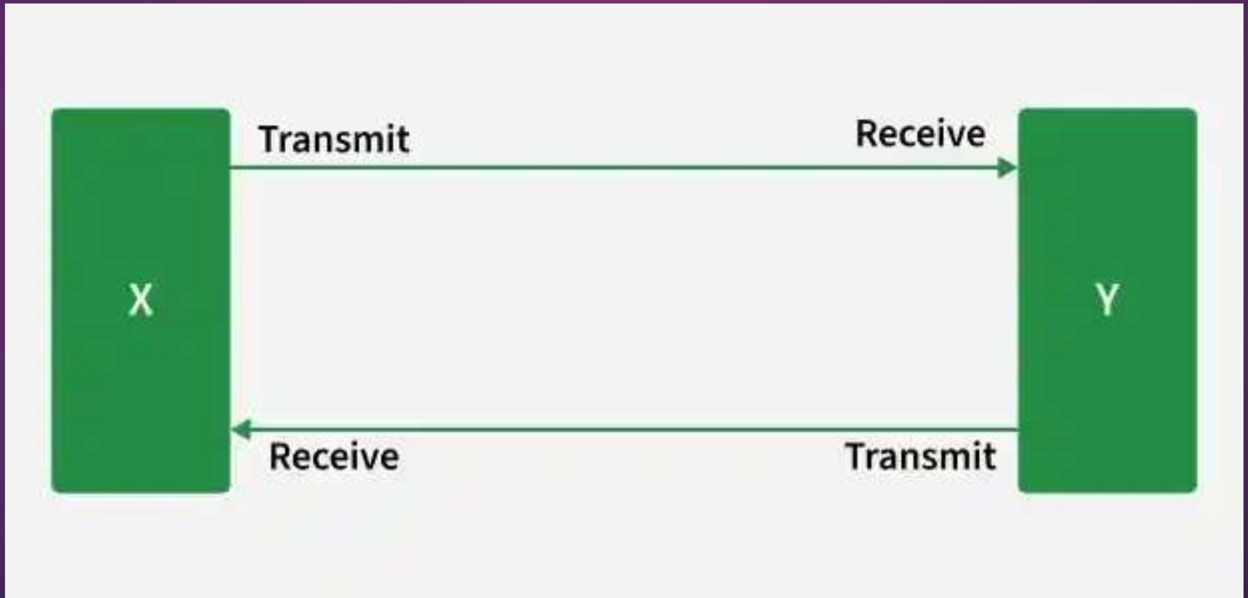
Network Topology

The network topology defines the relative configuration of different pieces of network equipment like cables, computers, and other peripherals. There are number of network topologies and a network could be build using multiple to pologies. Distinction is to be made between physical topology (with respect to lay out of the network) and logical topology (which defines information circulation at the lowest level) as it is very important to distinguish clearly between these two aspects.

main types of topology: Bus, Star, Ring, Mesh and Tree

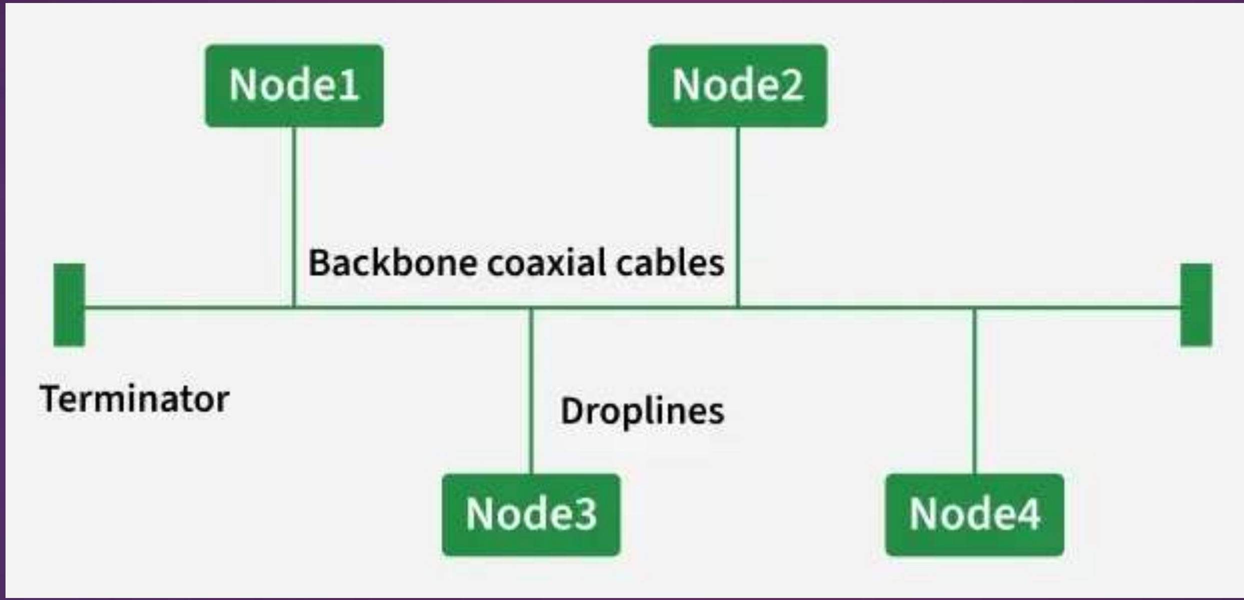
Point To Point Topology

- Point-to-point topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



i. Linear Bus Topology

A linear bus topology consists of a main run of cable with a terminator at each end . All nodes (file server, workstations, and peripherals) are connected to the linear cable.



Advantages

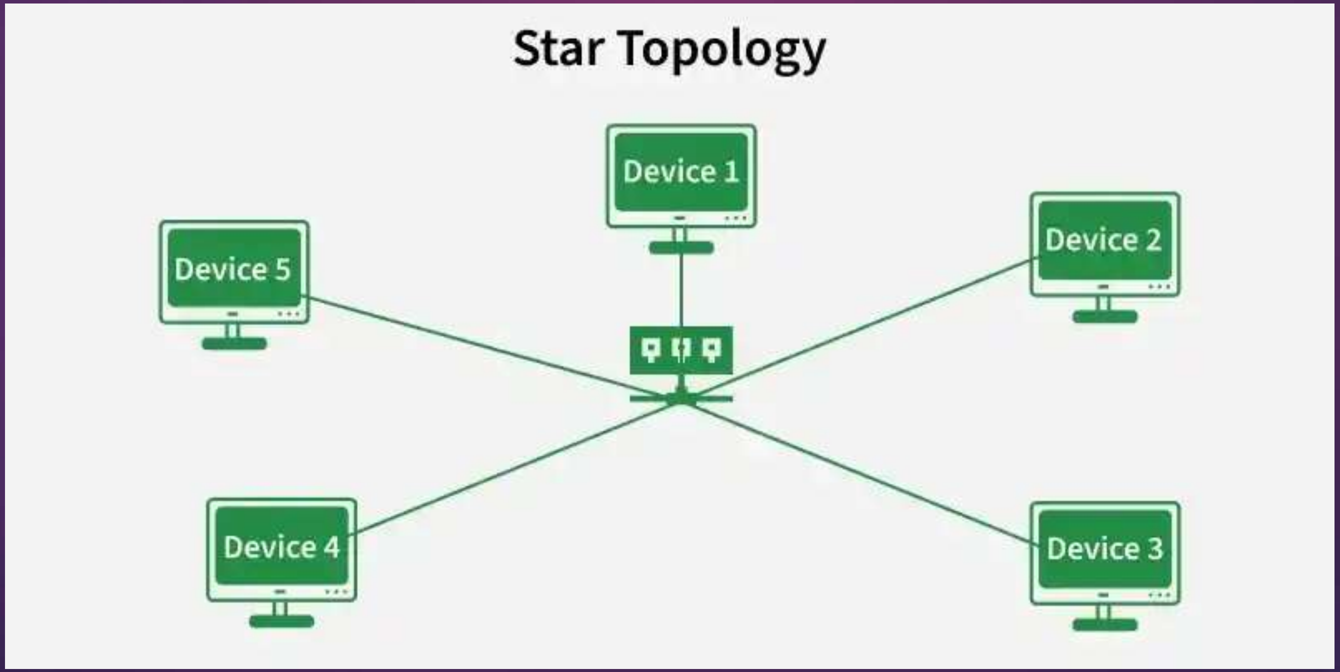
- Easy to connect a computer or peripheral to a linear bus.
- Requires less cable length than a star topology.

Disadvantages

- Entire network shuts down if there is a break in the main cable.
- Terminators are required at both ends of the backbone cable.
- Difficult to identify the problem if the entire network shuts down.
- Not meant to be used as a stand-alone solution in a large building

ii. Star Topology

A star topology is designed with each node (file server, workstations, and peripherals) connected directly to a central network hub, switch, or concentrator (See fig. 3). Data on a star network passes through the hub, switch, or concentrator before continuing to its destination. The hub, switch, or concentrator manages and controls all functions of the network. It also acts as a repeater for the data flow. This configuration is common with twisted pair cable; however, it can also be used with coaxial cable or fibre optic cable.



Advantages

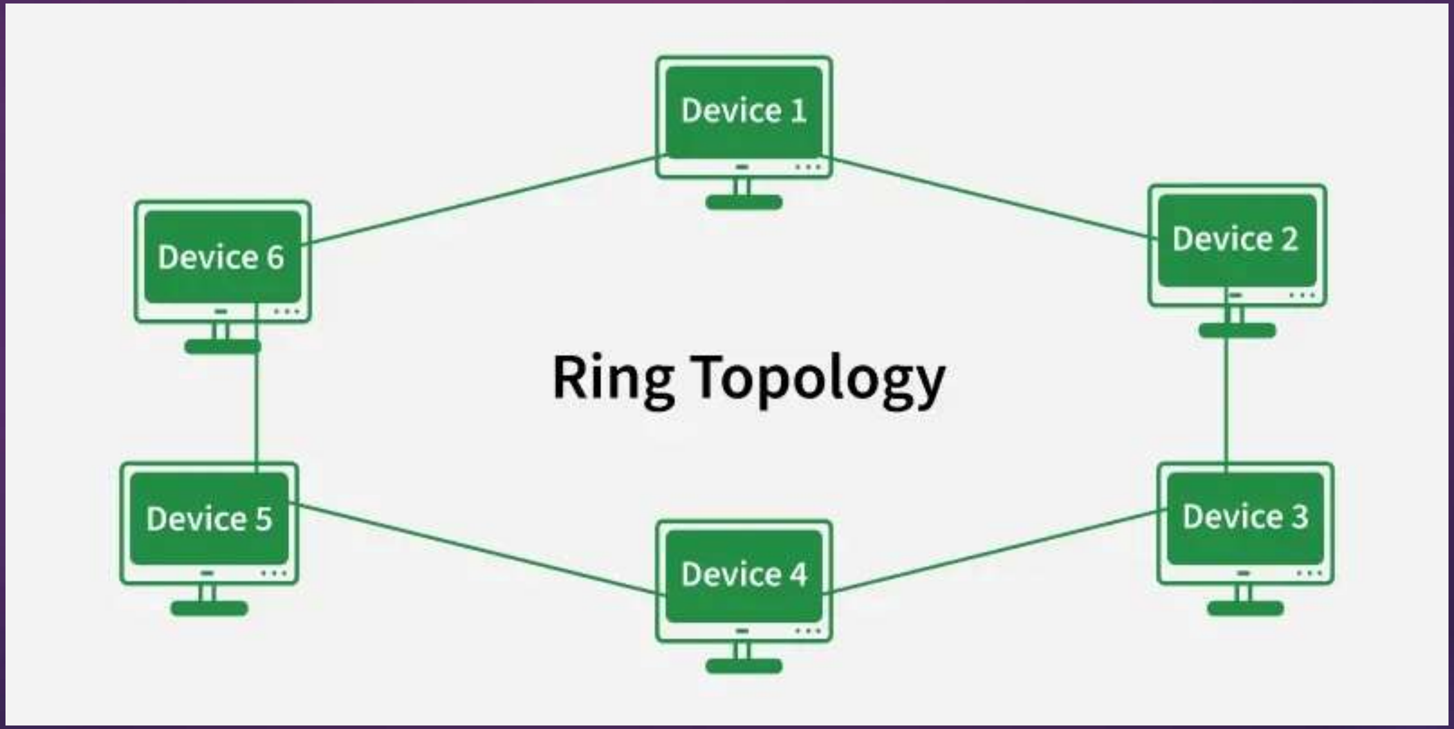
- Easy to install and wire.
- No disruptions to the network when connecting or removing devices.
- Easy to detect faults and to remove parts.

Disadvantages

- Requires more cable length than a linear topology.
- If the hub, switch, or concentrator fails, nodes attached are disabled.
- More expensive than linear bus topologies because of the cost of the hubs, etc.

iii. Ring Topology

In ring topology each node connects to exactly two other nodes, forming a single continuous pathway for signals through each node – a ring (see Fig 4). Data travels from node to node, with each node along the way handling every packet. Because a ring topology provides only one pathway between any two nodes, ring networks may be disrupted by the failure of a single link. A node failure or cable break might isolate every node attached to the ring.



Advantages

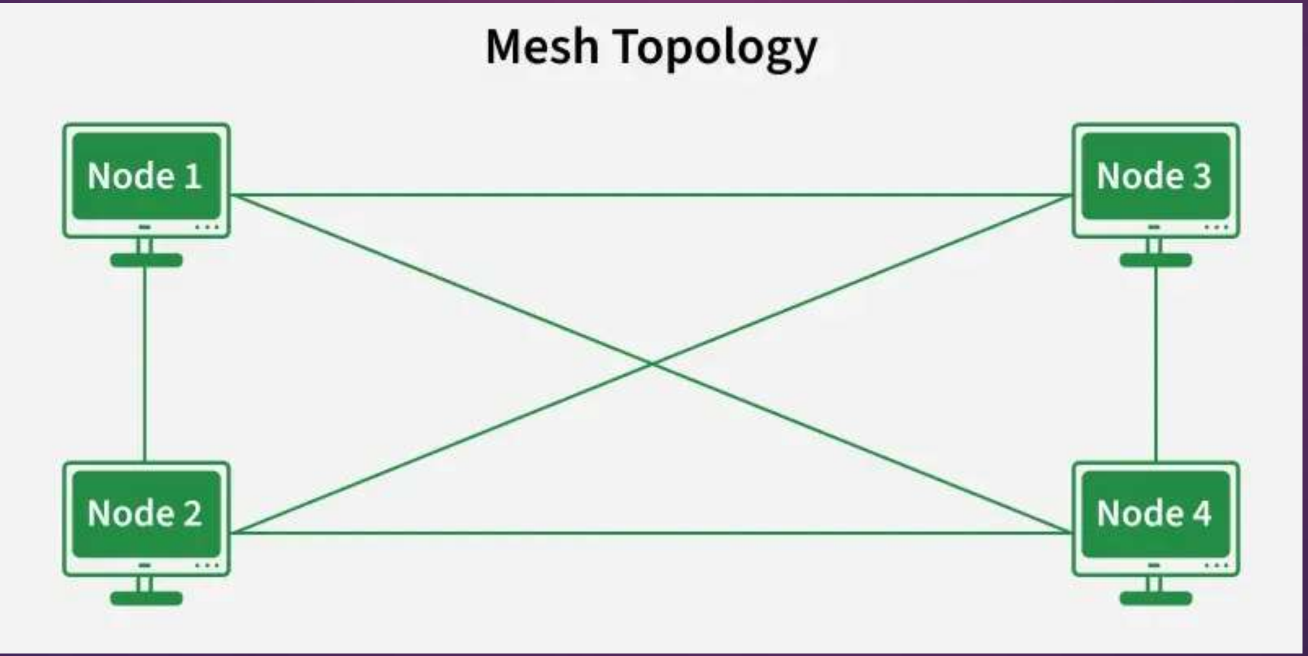
- Very orderly network where every device has access to the token and the opportunity to transmit.
- Performs better than a star topology under heavy network load.
- Can create much larger network using Token Ring.
- Does not require network server to manage the connectivity between the computers.
- Data is quickly transferred without a “bottle neck”. (very fast, all data traffic is in the same direction).
- The transmission of data is relatively simple as packets travel in one direction only.

Disadvantages

- One malfunctioning workstation or bad port in the MAU can create problems for the entire network.
- Data packets must pass through every computer between the senders and recipient therefore this makes it slower.
- Moves, adds and changes of devices can affect the network.
- Because all stations are wired together, to add a station you must shut down the network temporarily.
- Difficult to troubleshoot a ring network.
- In order for all computers to communicate with each other, all computers must be turned on (Network Troubleshooting and Resource Site for School IT Staff, n.d.).

iv. Mesh Topology

- In a mesh topology, every device is connected to another device via a particular channel. Every device is connected to another via dedicated channels. These channels are known as links. Mesh topology does not depend on specific protocols; it focuses on direct device-to-device connections.



- Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is $N - 1$.

- The total number of ports required = **N ∗ (N−1)**

- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is $\boldsymbol{{}^{N}C_{2}}$

- = **N ∗ (N−1)/2**

- 6 Device ?

Advantages

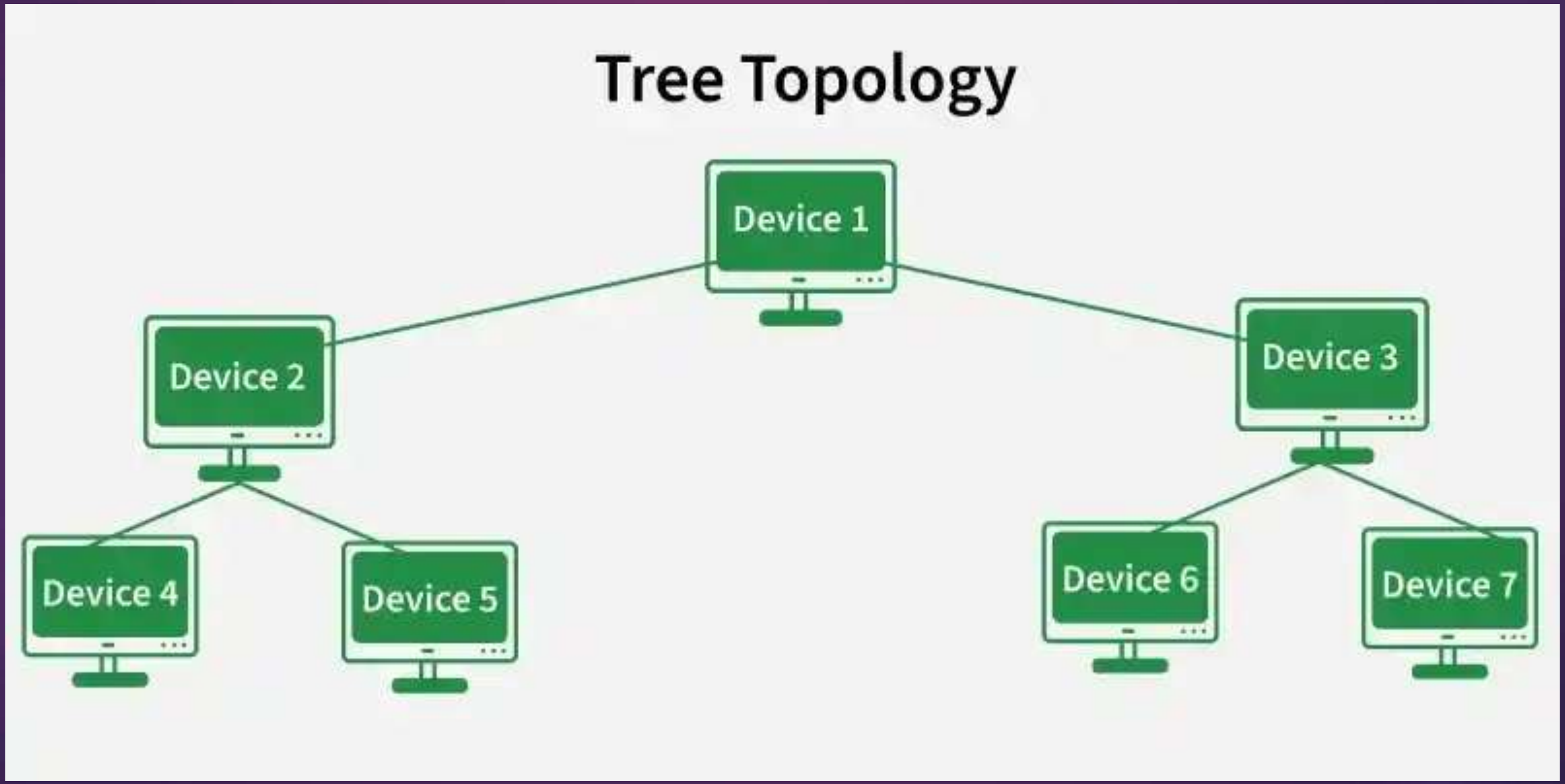
- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Disadvantages

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

v. Tree Topology

A tree topology combines characteristics of linear bus and star topologies. It consists of groups of star-configured workstations connected to a linear bus backbone cable (See fig. 5). Tree topologies allow for the expansion of an existing network, and enable schools to configure a network to meet their needs.



Advantages

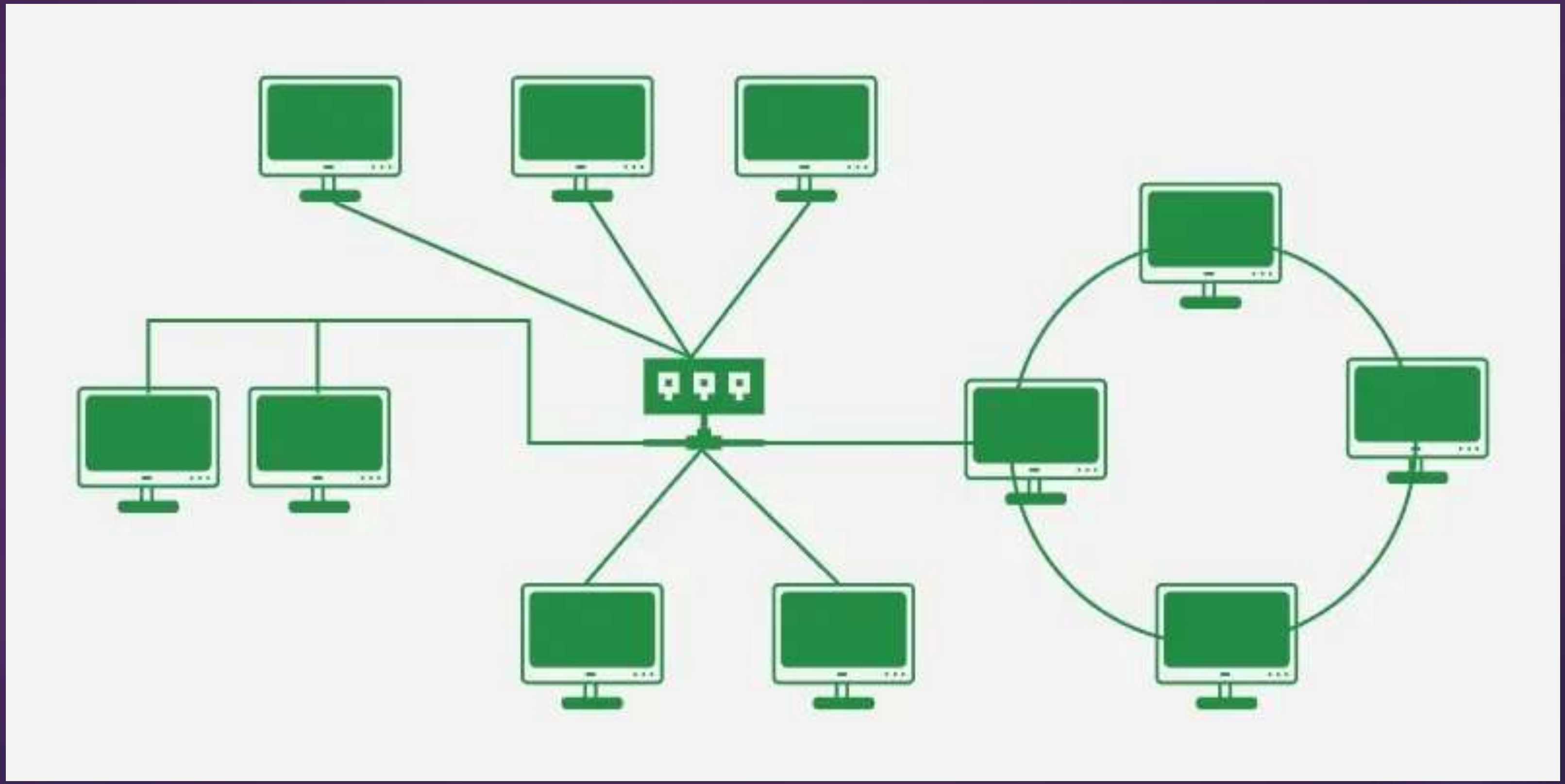
- Point-to-point wiring for individual segments.
- Supported by several hardware and software vendors.

Disadvantages

- Overall length of each segment is limited by the type of cabling used.
- If the backbone line breaks, the entire segment goes down.
- More difficult to configure and wire than other topologies (Florida Center for Instructional Technology. n.d.).

Hybrid Topology

- Hybrid Topology is the combination of all the various types of topologies we have studied above. Hybrid Topology is used when the nodes are free to take any form.



Advantages

- This topology is very flexible.
- The size of the network can be easily expanded by adding new devices.

Disadvantages

- It is challenging to design the architecture of the Hybrid Network.
- Hubs used in this topology are very expensive.
- The infrastructure cost is very high as a hybrid network requires a lot of cabling and network devices .

Important Points

Network Topology is important because it defines how devices are connected and how they communicate in the network. Here are some points that defines why network topology is important.

- **Network Performance:** Upon choosing the appropriate topology as per requirement, it helps in running the network easily and hence increases network performance.
- **Network Reliability:** Some topologies like Star, Mesh are reliable as if one connection fails, they provide an alternative for that connection, hence it works as a backup.
- **Network Expansion** : Choosing correct topology helps in easier expansion of Network as it helps in adding more devices to the network without disrupting the actual network.
- **Network Security:** Network Topology helps in understanding how devices are connected and hence provides a better security to the network.

If there are 'N' devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively.

Which Topology ?

Need for Computer Networks

Interoperability is the ability of diverse computers from different vendors and with different operating systems to cooperate in solving computational problems.

users use networks without knowing the details of hardware, software and communication methods involved in them. Internet is the best example that is now widely being used all over the world including India

availability of powerful hardware, software and networking technology facilitate fast and reliable information exchange between institutions as well as support resource sharing.

Advantages of Computer Networks

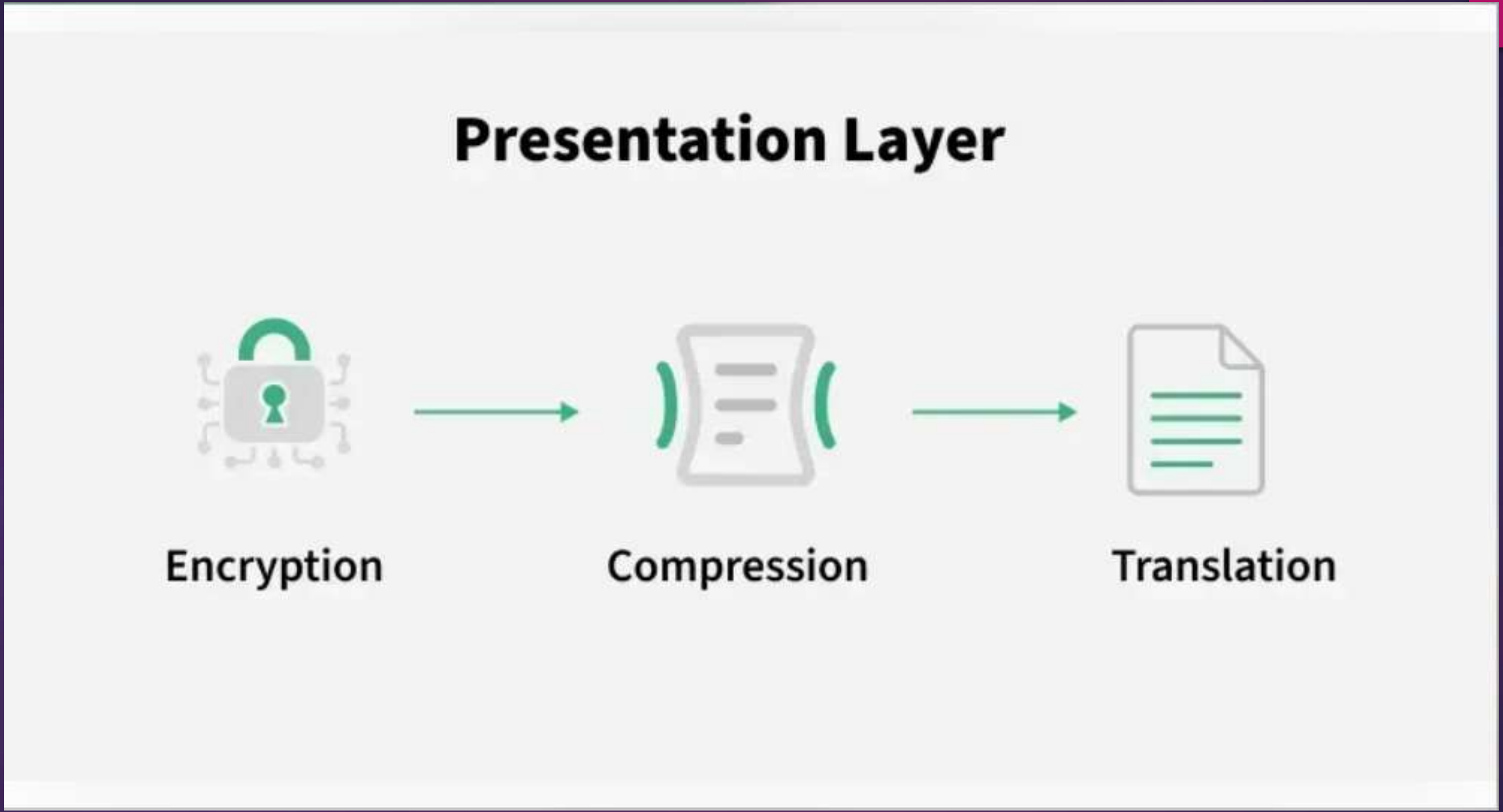
- File Transferring and Sharing Resources
- Speed
- User and Workgroup communication
- Increased Storage capacity
- Remote access
- Centralized software management

Layers of OSI Model

The OSI Model is a conceptual framework created by the International Organization for Standardization (ISO) to describe how data is transmitted across a network using a structured seven-layer architecture.

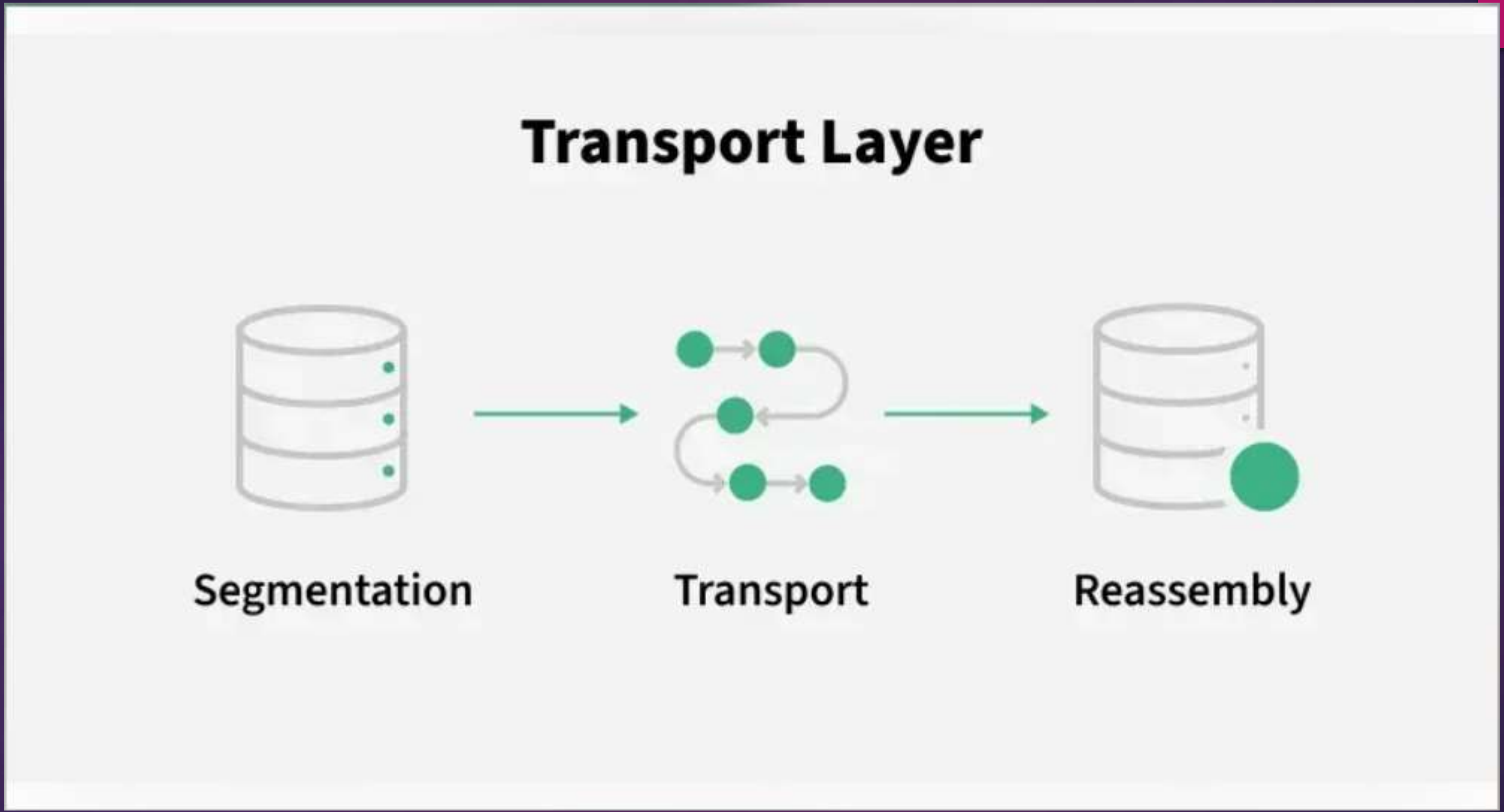
- Divides network communication into seven functional layers.
- Assigns specific responsibilities to each layer.
- Promotes compatibility between different networking systems.
- Simplifies network design, implementation, and troubleshooting.

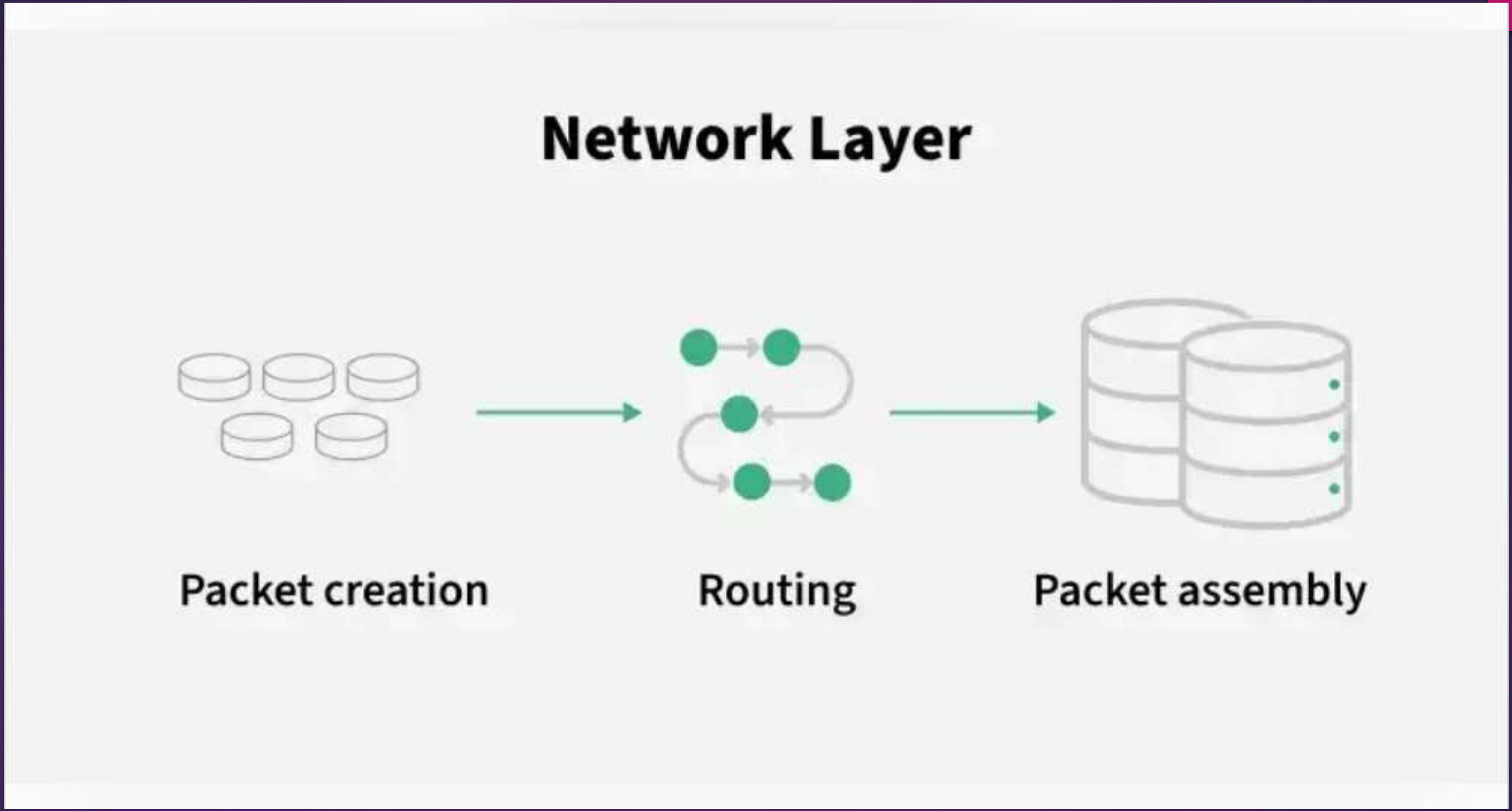
Application Layer

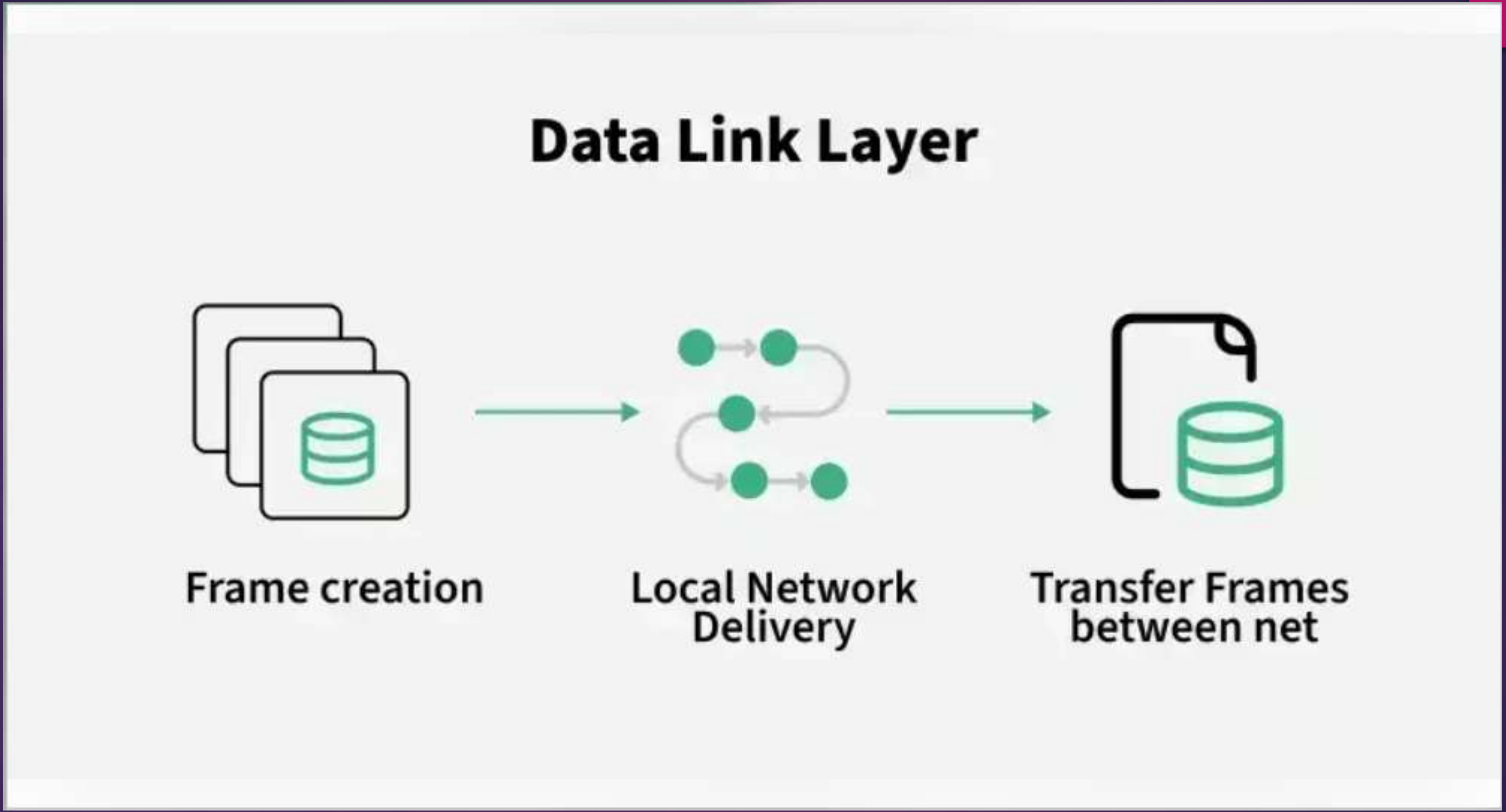

Presentation Layer
Encryption
Compression
Translation

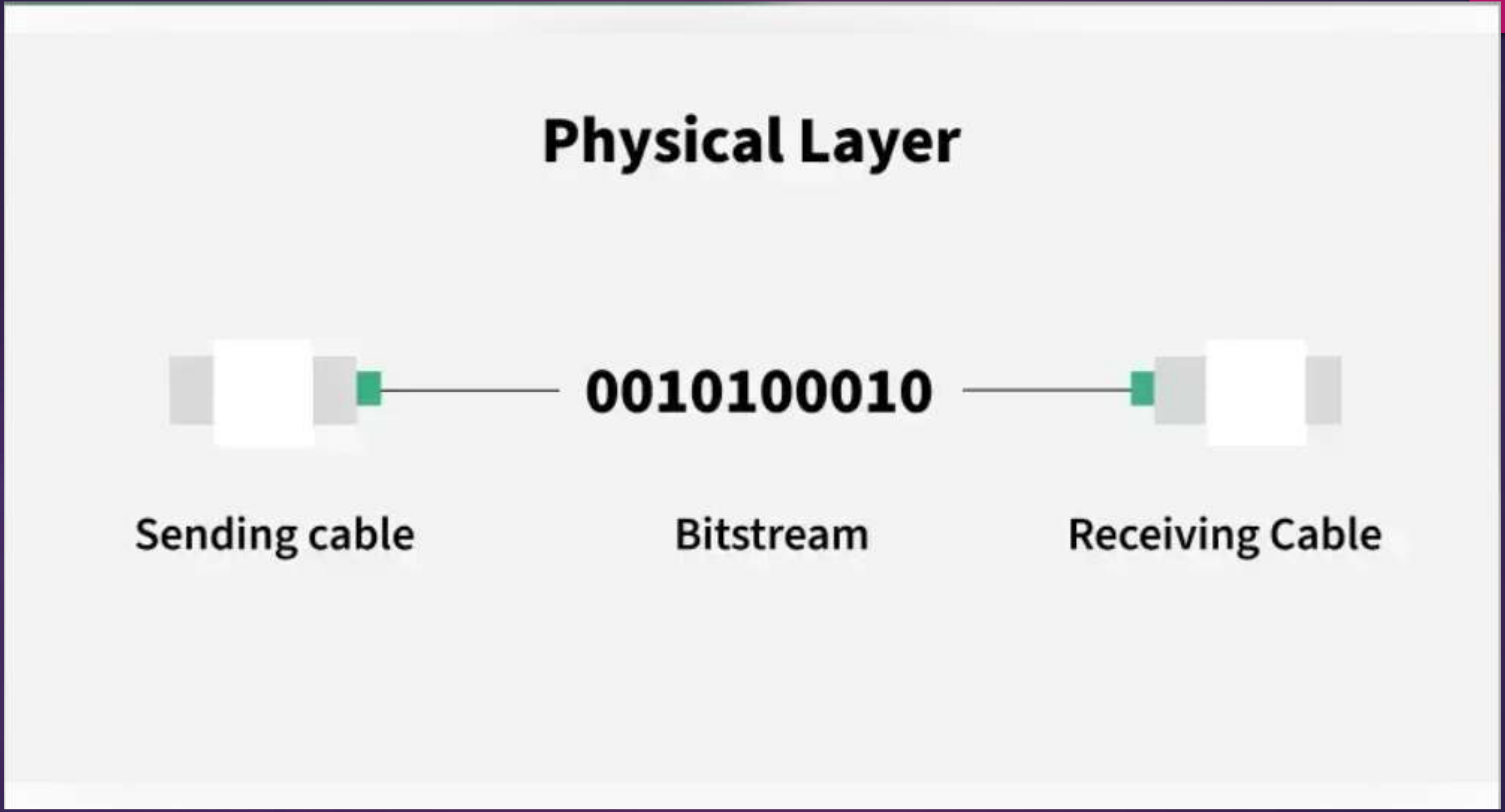
Session Layer

Session of Communication


Transport Layer
Segmentation
Transport
Reassembly


Network Layer
Packet creation
Routing
Packet assembly


Data Link Layer
Frame creation
Local Network Delivery
Transfer Frames between net


Physical Layer
0010100010
Sending cable
Bitstream
Receiving Cable